

Crystal Optics: Monochromators, analyzers

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Crystal Optics

Outline:

- Introduction: materials, dynamical diffraction
- Double Crystal Monochromators
- High Resolution mountings
- Curved crystals
- Analyzers/spectrometers: Johann, Johansson
- Laue crystals

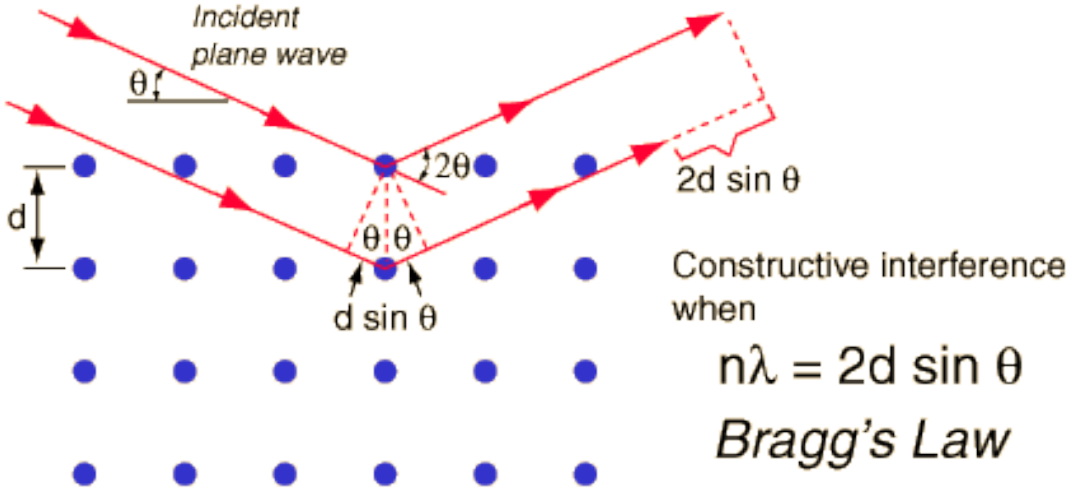
Historical Background

Three founding experiments

- Rontgen (1895): Discovery of X-rays (wave vs. corpuscle) unknown for 17 years
- Von Laue / Friedrich & Knipping (1912): ZnS crystal ⇒ periodic 3-D grating; confirmed electromagnetic wave nature and crystal periodicity
- W. L. & W. H. Bragg (1913): Specular reflection from lattice planes; first crystal spectrometer — direct ancestor of every monochromator

Diffraction in crystals

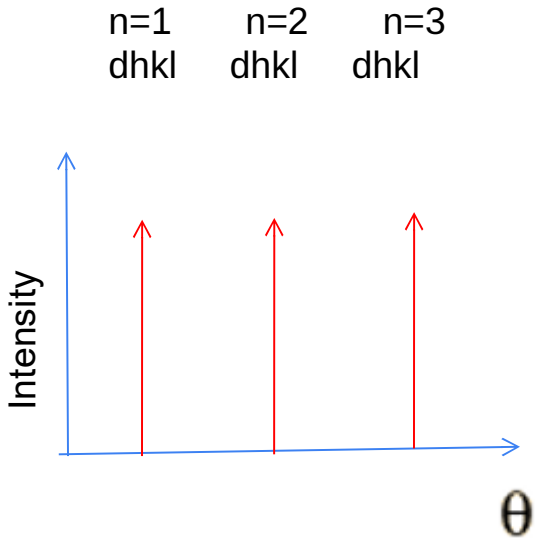
- A result of COHERENT (RAYLEIGH) scattering of the X-rays on the elements of a periodic structure (e.g., atoms).
- Although σ_R is small compared to other processes, this effect is the basis of X-ray diffraction.
- The (small) scattering is enhanced by the periodic distribution of the scatterers (atoms)



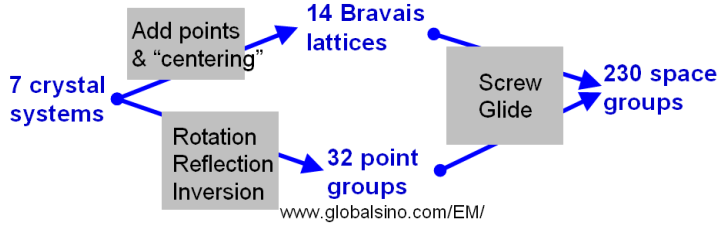
Constructive interference when

$$n\lambda = 2d \sin\theta$$

Bragg's Law



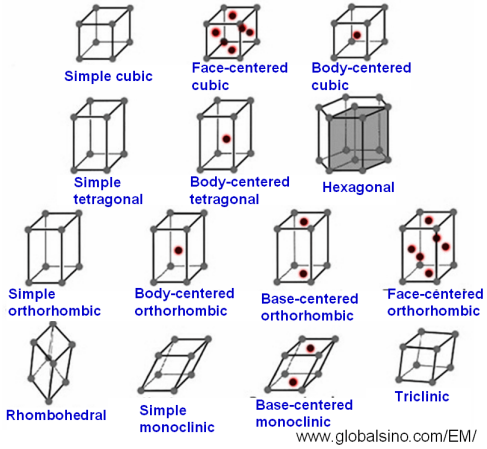
Crystal Structure



Crystal System Table

System	Axial length	Axial Angle	Unit Cell Geometry
Cubic	a = b = c	α = β = γ = 90°	
Tetragonal	a = b ≠ c	α = β = γ = 90°	
Orthorhombic	a ≠ b ≠ c	α = β = γ = 90°	
Rhombohedral	a = b = c	α = β = γ ≠ 90°	
Hexagonal	a = b ≠ c	α = β = 90°, γ=120°	
Monoclinic	a ≠ b ≠ c	α = γ = 90°, β≠90°	
Triclinic	a ≠ b ≠ c	α ≠ β ≠ γ	

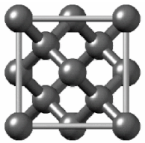
Bravais Lattices



https://www.globalsino.com/EM/page4546.html

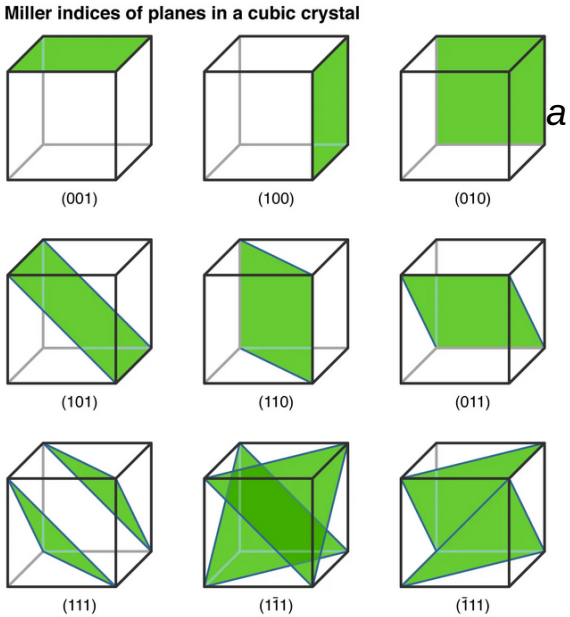
Crystal system	Crystal family		Required symmetries of point group	Point group	Space group	Bravais lattices			Pearson symbol	Coordinate description
		Symbol				No.	Symbol	Name		
Cubic	C		Four 3-fold rotation axes along a+b+c, -a+b+c, a-b+c, -a-b+c.	5	36	3	cP	Primitive cubic (P)	cP*	a=b=c, α=β=γ=90°
							cF	Face centered cubic (F)	cF*	
							cI	Body centered cubic (I)	cI*	

Diamond's cubic structure is in the Fd3m space group (space group 227), which follows the face-centered cubic Bravais lattice.
The diamond lattice can be viewed as a pair of intersecting face-centered cubic lattices, with each separated by 1/4 of the width of the unit cell in each dimension.



```
#S 6 Diamond
#UCELL 3.56679    3.56679    3.56679    90.0000    90.0000 90.0000
#USYSTEM Cubic
#ULATTICE F
#USTRUCTURE ZincBlende
#UREF R.W.G. Wyckoff, Crystal Structures, Interscience Publ. 1965, 1, p.26
#UTEMP 293.15
#UCOMMENT The ZincBlende structure is defined by atom A located at (0 0 0)
#UCOMMENT and atom B at (1/4 1/4 1/4) of the fcc lattice. Ex: Si(a=5.4309)
#UCOMMENT Ge(a=5.657)  Diamond(a=3.56)  GaAs  GaP  InAs InP  InSb  SiC...
#N  5
#L  AtomicNumber  Fraction  X  Y  Z
6 1.0 0.0  0.0  0.0
6 1.0 0.0  0.5  0.5
6 1.0 0.5  0.0  0.5
6 1.0 0.5  0.5  0.0
6 1.0 .25  .25  .25
6 1.0 .25  .75  .75
6 1.0 .75  .25  .75
6 1.0 .75  .75  .25
```

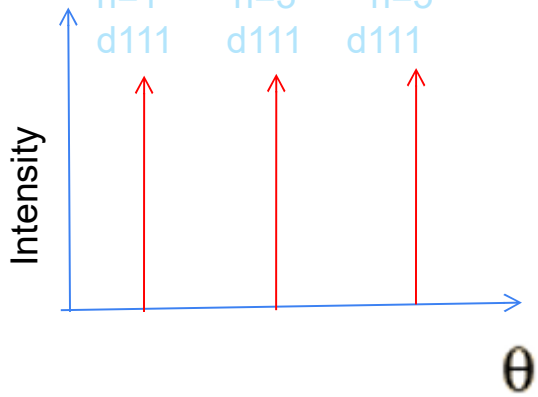
Crystal Structure



$$2\,d_{hkl}\,\sin\theta_B = n\lambda$$

n=1 d111 | n=1 d333 | n=1 d555

n=1 d111 | n=3 d111 | n=5 d111



$$F(h,k,l) = \sum_{j}^{n} C_j T_j f_j \exp\left[2\pi i\left(hx_j + ky_j + lz_j\right)\right]$$

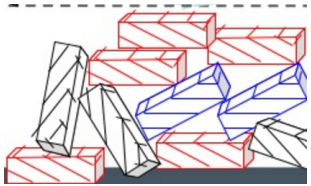
There are forbidden reflections: e.g. Si222

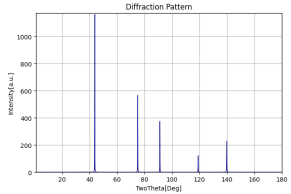
Type of crystals

$$R_{\mathrm{int}} = \frac{r_e^2\,\lambda^3\,|F_{hkl}|^2}{V_c^2\,\sin 2\theta_{\mathrm{B}}}$$

Powder

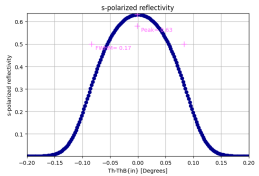
- Quasi-transparent grains randomly oriented in 3D
- Diffractogram: peaks at different dhkl
- Intensity: proportional to |Fhkl|^2
- Kinematic theory (very small crystals): single scattering: oscillators in the crystal are only under the influence of the incident wave





Mosaic

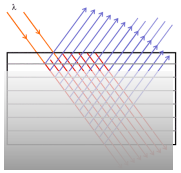
- “Oriented powder”
- Grains orientated Gaussian distribution around a main hkl
- “Large width” “Small peak reflectivity” “Secondary Extinction”

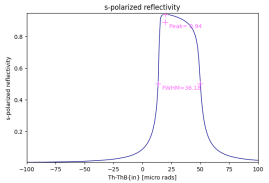


Perfect

- Super good crystals: No imperfection, dislocation, faults, etc
- Very big
- Primary extinction (surface)
- Dynamical theory: multiple scattering of the radiation emitted by the oscillators and its interaction with the incident wave

$$2\,d_{hkl}\,\sin\theta_{\mathrm{B}} = n\lambda \qquad \Delta\theta_{\mathrm{DW}} = \frac{4\,r_e\,\lambda^2\,|F_{hkl}|}{\pi\,V_c\,\sin 2\theta_{\mathrm{B}}}$$



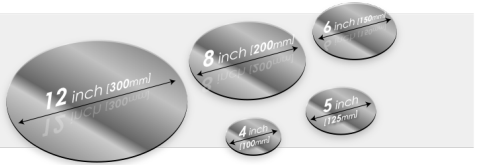


Materials and preparation

Si

- Gift from semiconductor industry
- The reference crystal





$\alpha = 2.56 \times 10^{-6}\,\mathrm{K}^{-1}$;
$\kappa = 148\,\mathrm{W\,m^{-1}\,K^{-1}}$

$$d_{hkl} = \frac{5.43\ \text{Å}}{\sqrt{h^2+k^2+l^2}}$$

Ge

- Wider Darwin width (×2)
- Higher d-spacing
- Less perfect
- More expensive

$$d_{hkl} = \frac{5.66\ \text{Å}}{\sqrt{h^2+k^2+l^2}}$$

Diamond

- Less perfect
- Essential for high-heat-load undulator beamline $\kappa = 2200\,\mathrm{W\,m^{-1}\,K^{-1}}$
- Limited size

$$d_{hkl} = \frac{3.56\ \text{Å}}{\sqrt{h^2+k^2+l^2}}$$

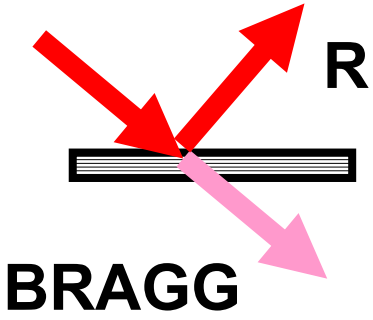
Others

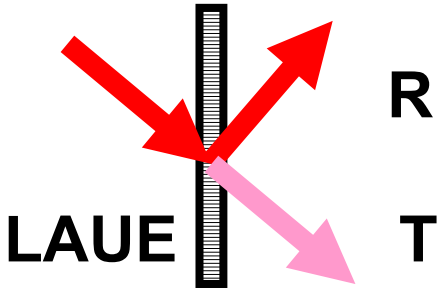
- Quarts, LiF, etc
- High d-dpacing

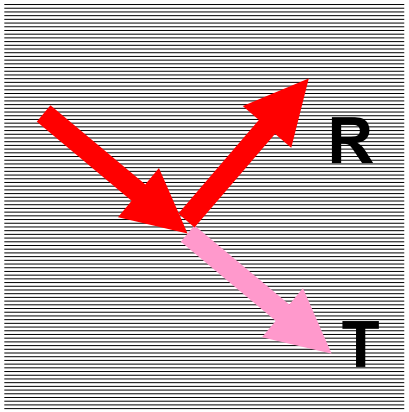
They must be tailored into monochromators etc.

- Orientation, cutting, lapping, polishing and etching.
- Strain free crystal preparation.
- Accurate and stable mounting.
- Adequate cooling scheme.

Bragg & Laue crystal geometries

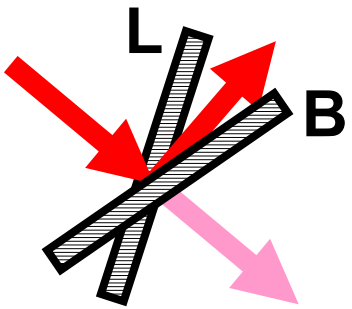






Transition:

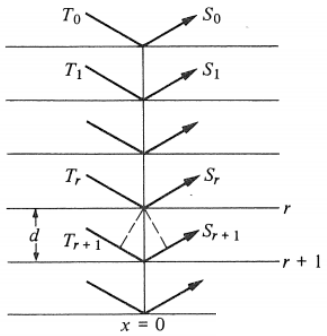
$$\theta_B = \alpha$$



DYNAMICAL THEORY (1914) – the Darwin width

Darwin, C. G. (1914). Philos. Mag., 27, 315-333. The Theory of X-ray Reflection.
Darwin, C. G. (1914). Philos. Mag., 27, 675-690. The Theory of X-ray Reflection. Part II.

DARWIN TREATM



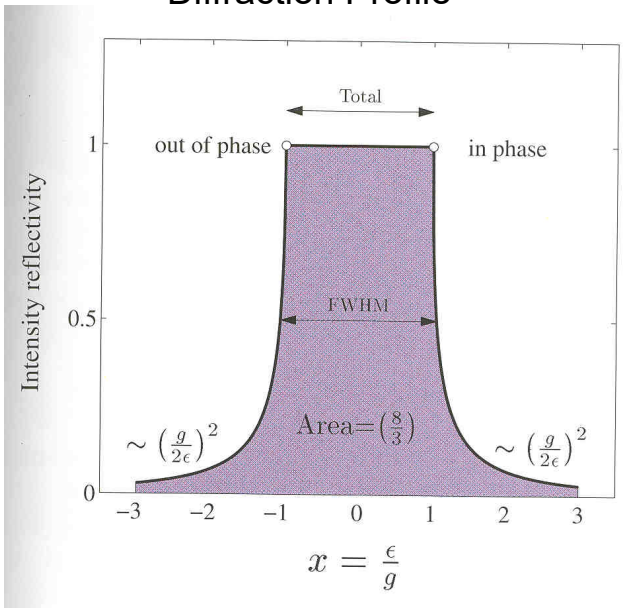
$$S_r = \frac{T_r - (1 - h + iq_0)T_{r-1}e^{-i\phi}}{iqe^{-i2\phi}}$$

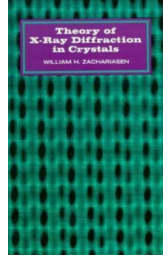
$$R(x) \equiv \begin{cases} \left(x - \sqrt{x^2 - 1}\right)^2 & \text{for } x \geqq 1 \\ 1 & \text{for } |x| \leqq 1 \\ \left(x + \sqrt{x^2 - 1}\right)^2 & \text{for } x \leqq -1 \end{cases}$$

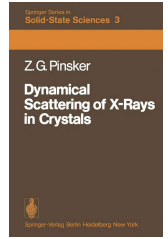
$$(\Delta\theta)_D = 2\left|\frac{P|\Psi_H|}{(|b|)^{1/2}\sin(2\theta_B)}\right|$$

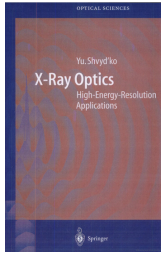
$$\Psi_H = \frac{-r_0\lambda^2}{\pi v_c}F_H, \qquad r_0 = \frac{e^2}{mc^2}$$

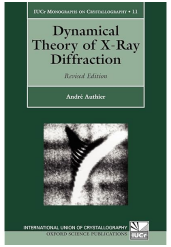
"Diffraction Profile"











Guigay & Sanchez del Rio (undistorted): crystalpy library

Guigay and Sanchez del Rio (2024) https://doi.org/10.1107/S160057752400924X

$$\frac{\partial D_0}{\partial s_0} = \frac{ik}{2}\left[\chi_0 D_0 + \chi_{-h} D_h\right] = iu_0 D_0 + iu_{-h} D_h, \qquad (10a)$$

$$\frac{\partial D_h}{\partial s_h} = \frac{ik}{2}\left[(\chi_0 + \alpha) D_h + \chi_h D_0\right] = i(u_0 + \alpha') D_h + iu_h D_0, \qquad (10b)$$

$$\begin{pmatrix} D_0(T) \\ D_h(T) \end{pmatrix} = M \begin{pmatrix} D_0(0) \\ D_h(0) \end{pmatrix} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} D_0(0) \\ D_h(0) \end{pmatrix}.$$

$$m_{11} = \left[\cos(aT) - i\frac{\omega}{a}\sin(aT)\right]\exp[iT(u_0+\omega)],$$

$$m_{12} = i\frac{u_{-h}}{a}\sin(aT)\exp[iT(u_0+\omega)],$$

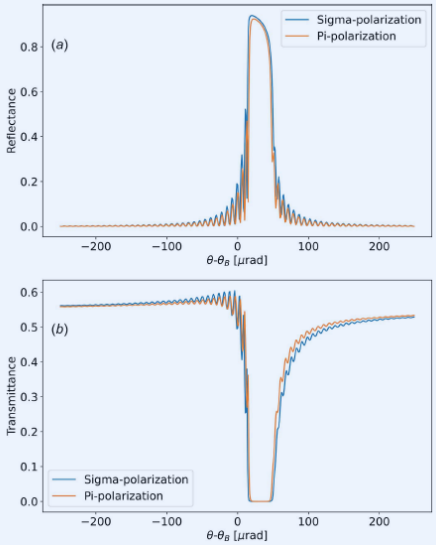
$$m_{21} = ib\frac{u_h}{a}\sin(aT)\exp[iT(u_0+\omega)],$$

$$m_{22} = \left[\cos(aT) + i\frac{\omega}{a}\sin(aT)\right]\exp[iT(u_0+\omega)].$$

$$r_B = -\frac{m_{21}}{m_{22}} = \frac{-ibu_h \sin(aT)}{a\cos(aT) + i\omega\sin(aT)}, \qquad (39a)$$

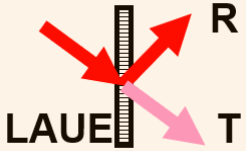
$$t_B = m_{11} + m_{12} r_B = \frac{a\exp[iT(u_0+\omega)]}{a\cos(aT) + i\omega\sin(aT)}. \qquad (39b)$$

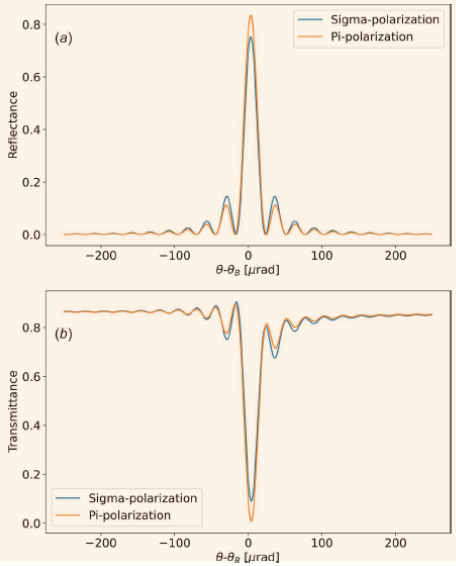




$$r_L = m_{21} = ibu_h \frac{\sin(aT)}{a}\exp[iT(u_0+\omega)], \qquad (34a)$$

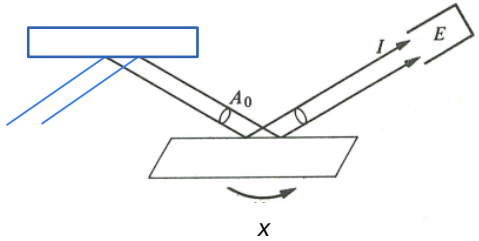
$$t_L = m_{11} = \left[\cos(aT) - i\omega\frac{\sin(aT)}{a}\right]\exp[iT(u_0+\omega)]. \qquad (34b)$$

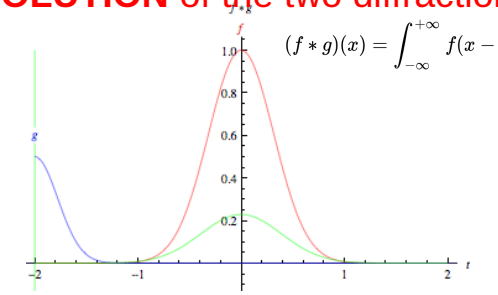




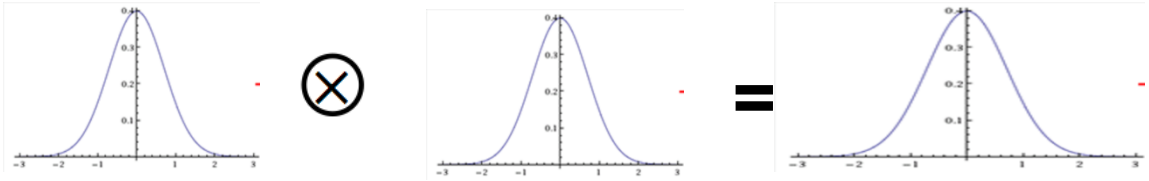
Rocking curves

The rocking curve is the **CONVOLUTION** of the two diffraction profiles.

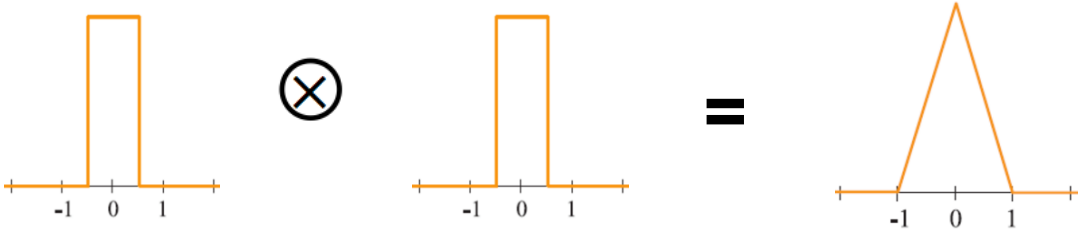




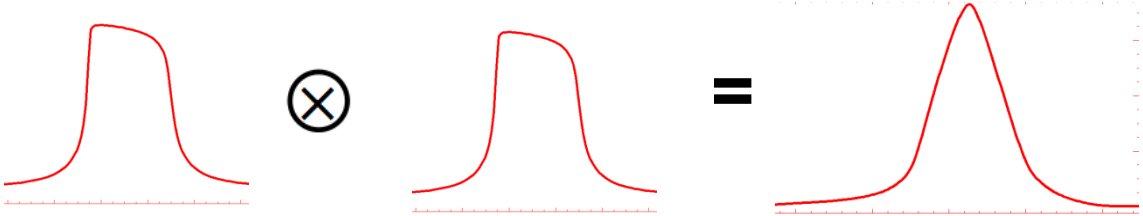
$$(f * g)(x) = \int_{-\infty}^{+\infty} f(x-t)g(t)\,\mathrm{d}t = \int_{-\infty}^{+\infty} f(t)g(x-t)\,\mathrm{d}t$$



$FWHM_R = 1.4\ FWHM_1$



$FWHM_R = 1\ FWHM_1$

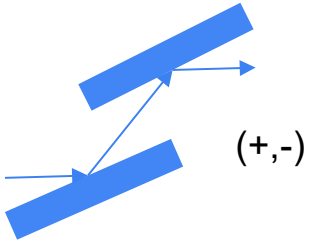


$FWHM_R \sim 1.5\ FWHM_1$

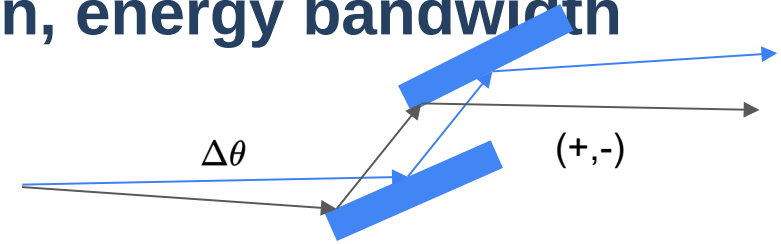
Si111 8 keV

For a full discussion about rocking curves see: Masiello *et al.* J. Appl. Crystall. 47, 1304-1314 (2014)

Two crystals: intrinsic resolution, energy bandwidth



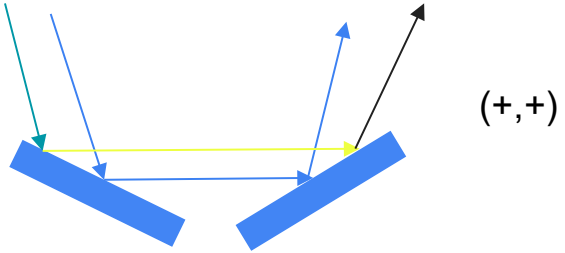
$$2\,d_{hkl}\,\sin\theta_B = n\lambda$$

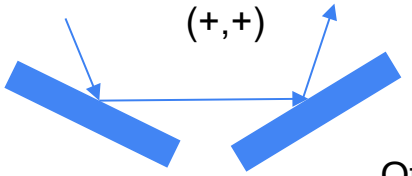


All rays accepted by the first crystal are also accepted (same angle) at the second crystal

$$\frac{\Delta E}{E} = \cot\theta_B \cdot \Delta\theta$$

$$\sqrt{(\Delta\theta)^2 + \omega^2}$$





Other facts:

- A parabolic mirror makes $\Delta\theta = 0$
- Source size term: $\Delta = \frac{s}{p}$

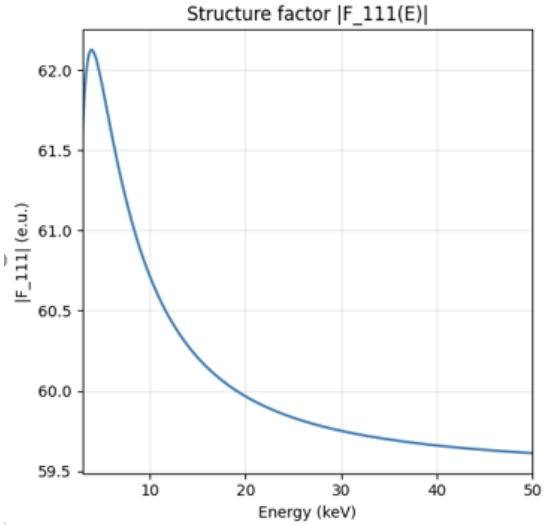
Out of axis rays on the first crystal are incident at different angles on the second crystal.

Intrinsic resolution is “almost constant” vs photon energy

$$\frac{\Delta E}{E} = \cot\theta_B \cdot \Delta\theta_{\mathrm{DW}}$$

$$\boxed{\frac{\Delta E}{E} = \frac{4\, r_e\, d_{hkl}^2\, |F_h|}{\pi\, V_c}}$$

$$\Delta\theta_{\mathrm{DW}} = \frac{2\, r_e\, \lambda^2\, |F_h|}{\pi\, V_c\, \sin 2\theta_B}$$



A “bit less” if

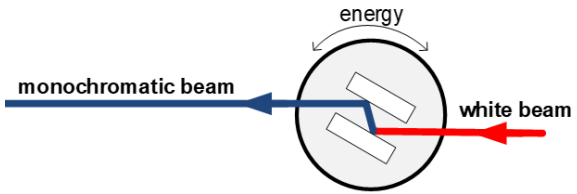
- Absorption
- Two crystals

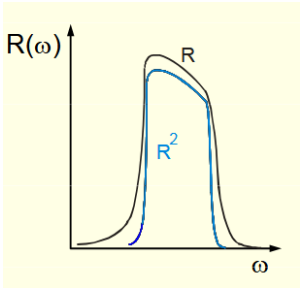
Therefore intrinsic resolution for Si111 $\frac{\Delta E}{E} \approx 10^{-4}$

E (keV)	θ_B (°)	$\lvert F_{111}\rvert$	Δθ_DW (")	ΔE/E (num)
3.0	41.23	61.57	24.51	1.356×10^{-4}
10.0	11.40	60.72	5.56	1.337×10^{-4}
50.0	2.27	59.61	1.07	1.313×10^{-4}

DCM: Double crystal monochromatos

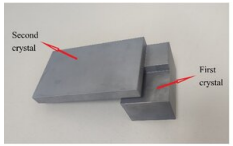
Fixed exit (approximately)
Non-dispersive (sensitive t $\Delta\theta$)
High harmonics



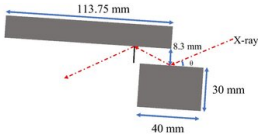


Channel cut:

- Simplicity (1 movement)
- Perfect alignment
- Vibration immunity
- Misalignment by elasticity (weak links + piezo)
- NOT fixed exit

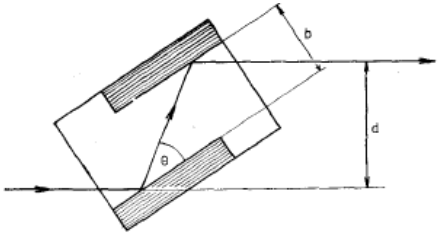


(a)



(b)

$$d = 2b\cos\theta$$



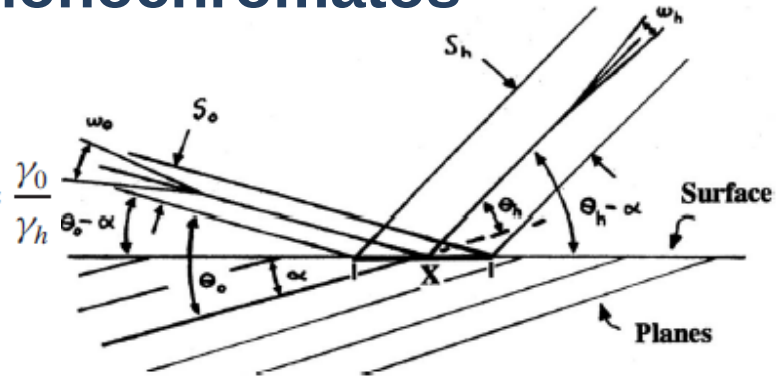
Separated crystals:

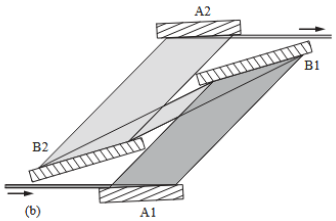
- More difficult alignment (< Darwin Width)
- More than 1 movement
- Flexibility (dimensions, geometry, cooling)
- Better crystals (polishing)

High Resolution crystal monochromatos

How to improve resolution (with flat crystals)

- Multiple crystals (product of diffraction pattern)
- Exploit asymmetry $\gamma_{0,h} = \mathbf{n}.\hat{\mathbf{s}}_{0,h} \equiv \cos(\theta_{0,h})$ $b = \frac{\gamma_0}{\gamma_h}$
- Dispersive setup (remove effects of divergenc
- High reflections (reduce Darwin width)
- Approach normal incidence (cot -> 0)





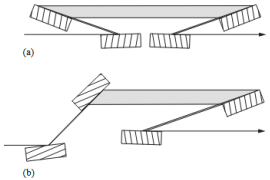
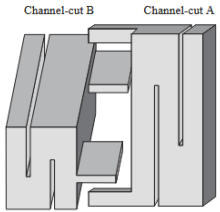


Fig. 8. Ultrahigh resolution monochromator. $(+m,-m,-m,+m)$ arrangement for higher resolution (a) and $(+n,-n,-m,+m)$ arrangement for higher throughput.



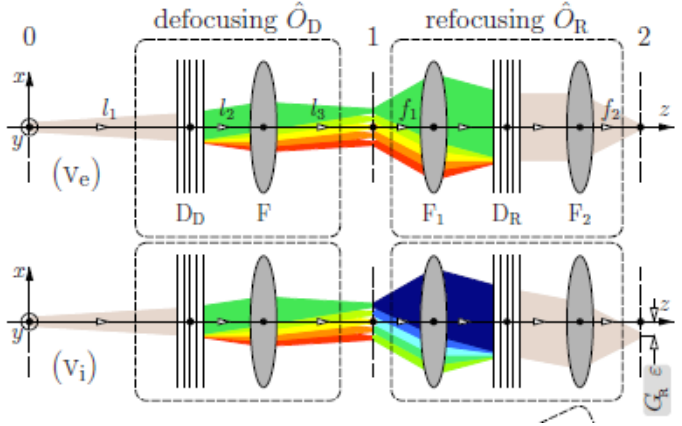
Reflection	$\Delta E/E$	Energy range
Si 111	1.1×10^{-4}	$\geqslant$1.98 keV
Si 220	5.8×10^{-5}	$\geqslant$3.23 keV
Si 311	2.8×10^{-5}	$\geqslant$3.79 keV
Si 331	1.4×10^{-5}	$\geqslant$4.98 keV
Si 422	1.4×10^{-5}	$\geqslant$5.59 keV
Si 333	8.5×10^{-6}	$\geqslant$5.93 keV
Si 440	9.3×10^{-6}	$\geqslant$6.46 keV
Si 620	6.5×10^{-6}	$\geqslant$7.22 keV
Si 444	4.8×10^{-6}	$\geqslant$7.91 keV
Si 555	1.5×10^{-6}	$\geqslant$9.89 keV
Si 777	3.5×10^{-7}	$\geqslant$13.84 keV

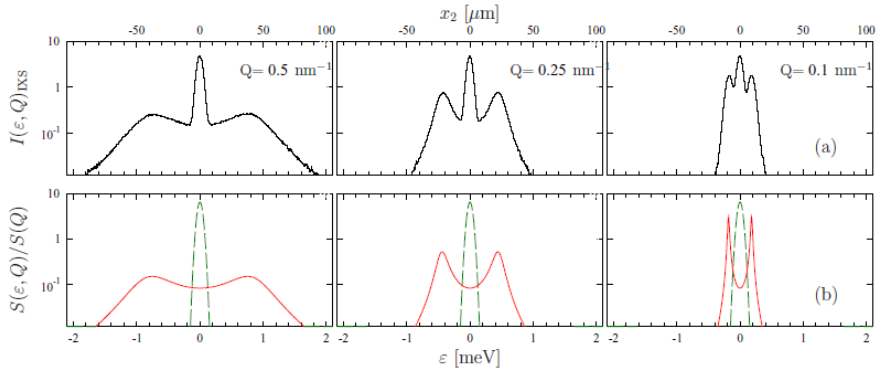
https://doi.org/10.1016/j.nima.2005.05.010

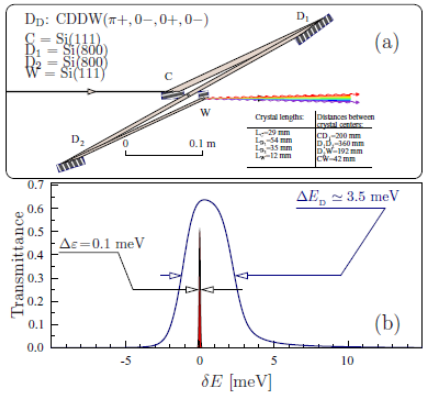
e.g.: X-ray echo spectrometers

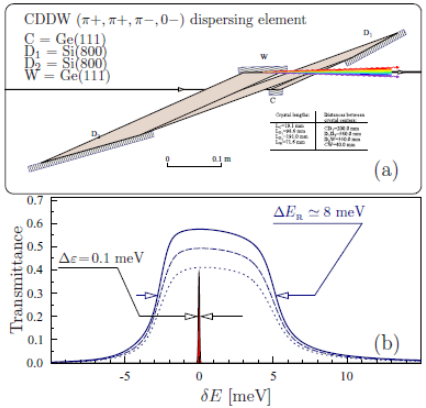
Defocusing: Dispersive monochromator
- Dispersive system (4 crystals)
- Focusing system (Lenses, KB or Montel)

Refocusing: Analyzer
- Dispersive system (4 crystals)
- Focusing system (Lenses, KB or Montel)



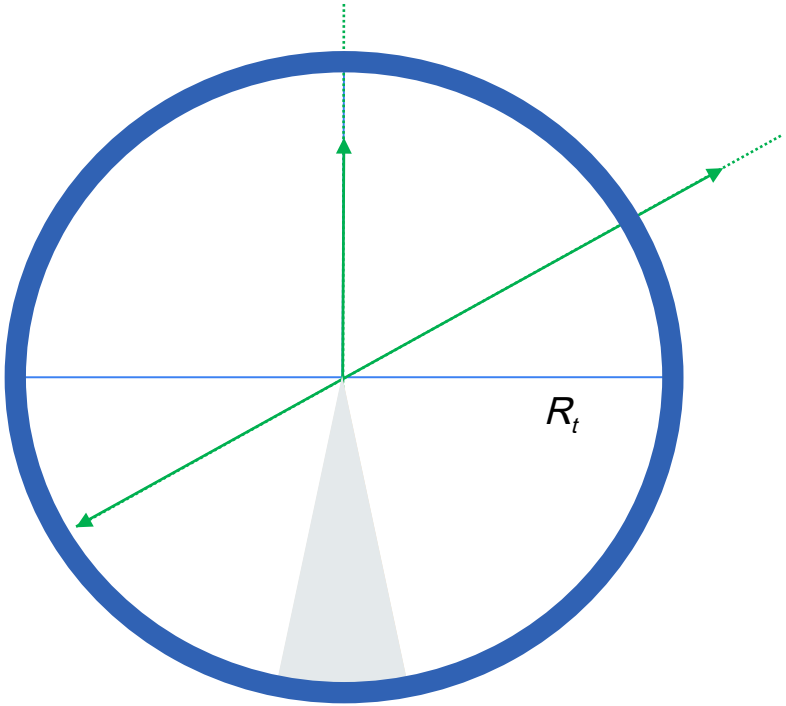




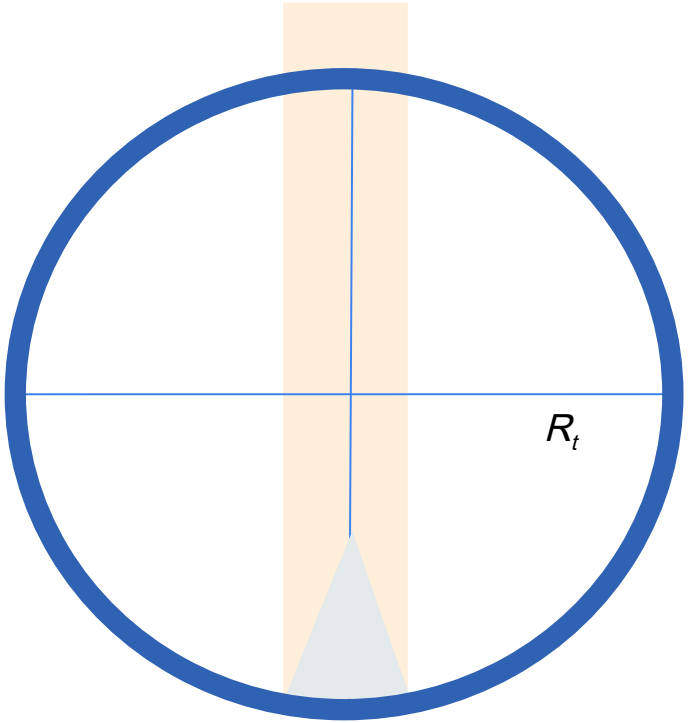


M Sánchez del Río, Y Shvyd'ko (2019) Phys. Rev. A 99
http://dx.doi.org/10.1103/PhysRevA.99.053839

Curved crystals: Bragg-Symmetric



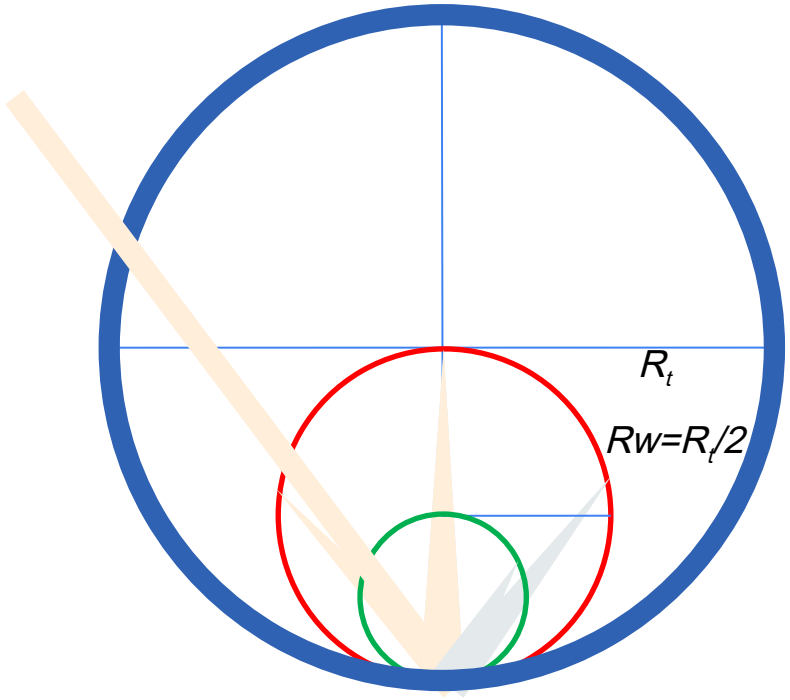
Curved crystals: Bragg-Symmetric



$$\frac{1}{p}+\frac{1}{q}=\frac{2}{R_t \sin\theta_B}$$

$$\theta_B = 90^\circ \rightarrow p = q = R_t;\ \ p = \infty, q = R_t/2$$

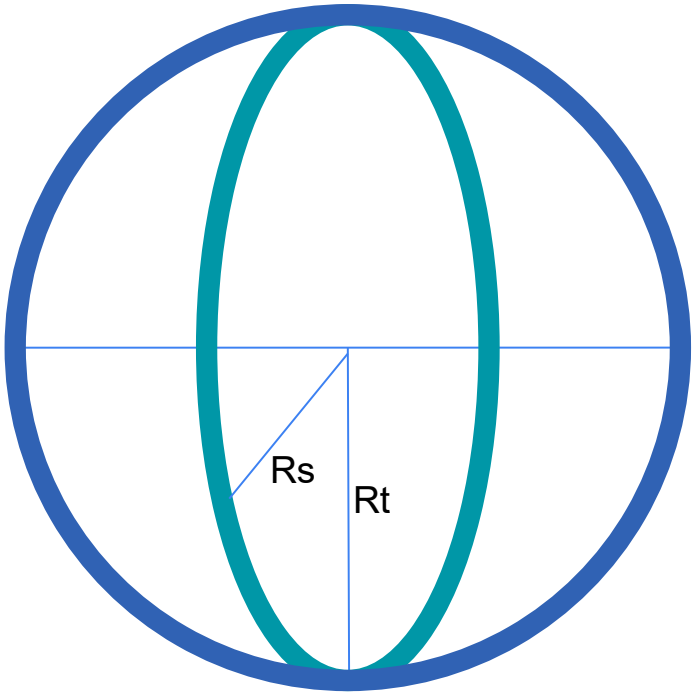
Curved crystals: Bragg-Symmetric



$$\frac{1}{p}+\frac{1}{q}=\frac{2}{R_t \sin\theta_B}$$

$$p=q=R_t\sin\theta_B;\quad p=\infty, q=(R_t/2)\sin\theta_B$$

Curved crystals: Bragg-Symmetric



$$\frac{1}{p}+\frac{1}{q}=\frac{2}{R_t \sin\theta_B}$$

$$\frac{1}{p}+\frac{1}{q}=\frac{2\sin\theta_B}{R_s}$$

$$\theta_B = 90^\circ \rightarrow R_t = R_s$$

Astigmatic if:

- Sphere in normal incidence
- Toroid (for a fixed Bragg angle)

Aberrations if q<<p

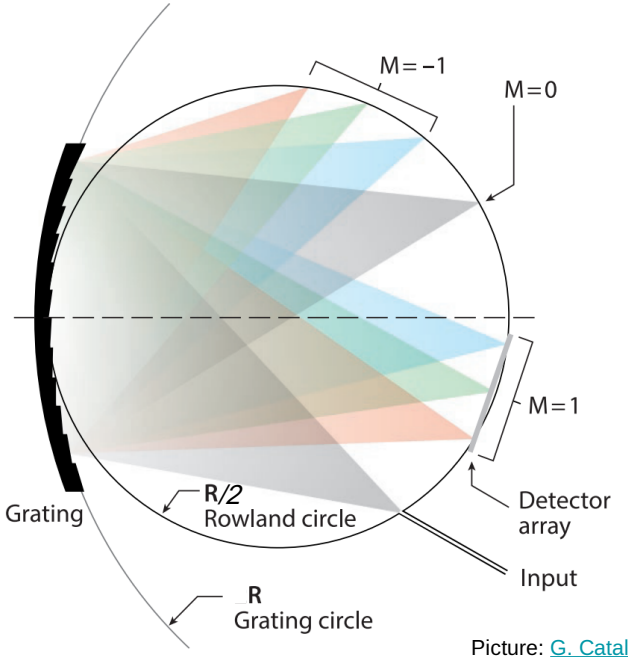
- Ellipse

Rowland circle

H. A. Rowland

"Preliminary notice of the results accomplished in the manufacture and theory of gratings for optical purposes" **Philosophical Magazine**, Series 5, **vol. 13**, pp. 469–474, **1882**.

Rowland, H. A. (1883). XXIX. On concave gratings for optical purposes. The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science, 16(99), 197–210. https://doi.org/10.1080/14786448308627419



Picture: G. Cataldo

$$\frac{1}{d} = k = \frac{\partial n}{\partial w} = \frac{1 + 2b_2 w + 3b_3 w^2 + \dots}{d_0} \quad (2)$$

$$F_{00} = r + r' \quad (3)$$

$$F_{10} = m\frac{\lambda}{d_0} - (\sin\alpha + \sin\beta) \quad (4)$$

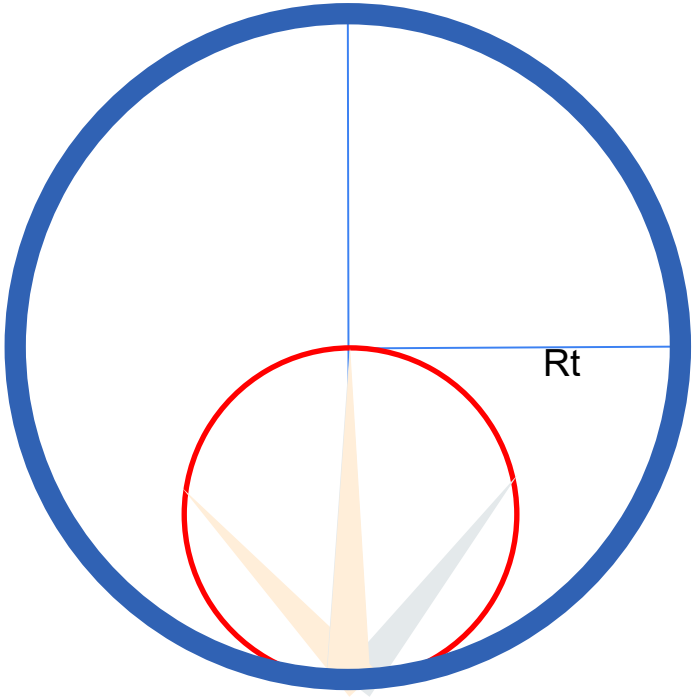
$$F_{20} = \frac{1}{2}\left(\frac{\cos^2\alpha}{r} - \frac{\cos\alpha}{R} + \frac{\cos^2\beta}{r'} - \frac{\cos\beta}{R}\right) - \frac{m\lambda b_2}{d_0} \quad (5)$$

$$F_{02} = \frac{1}{r} - \frac{\cos\alpha}{\rho} + \frac{1}{r'} - \frac{\cos\beta}{\rho} \quad (6)$$

$$F_{12} = \frac{1}{2}\left(\frac{\sin\alpha}{r}\left(\frac{1}{r} - \frac{\cos\alpha}{\rho}\right) + \frac{\sin\beta}{r'}\left(\frac{1}{r'} - \frac{\cos\beta}{\rho}\right)\right) \quad (7)$$

$$F_{30} = \frac{1}{2}\left(\frac{\sin\alpha}{r}\left(\frac{\cos^2\alpha}{r} - \frac{\cos\alpha}{R}\right) + \frac{\sin\beta}{r'}\left(\frac{\cos^2\beta}{r'} - \frac{\cos\beta}{R}\right)\right) - \frac{m\lambda b_3}{d_0} \quad (8)$$

Curved crystals: Bragg-Symmetric in Rowland circle configuration



$$\frac{1}{p}+\frac{1}{q}=\frac{2}{R_t \sin\theta_B}$$

$$p=q=R_t \sin\theta_B = 2R_w \sin\theta_B$$

Bragg symmetric

- p=q -> M=1

Rowland circle:

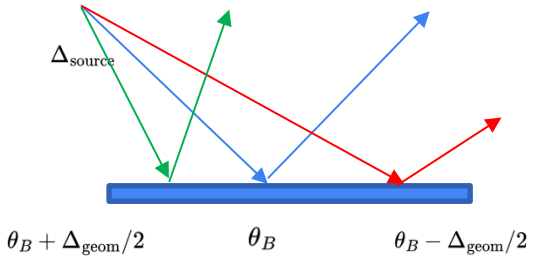
- Good focusing
- Not so good transmission (dispersion in θ_B)
- Best resolution

Curved crystals: Bragg-Symmetric in Rowland circle conf

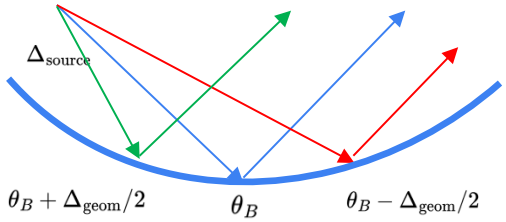
$$\frac{1}{p}+\frac{1}{q}=\frac{2}{R_t \sin\theta_B}$$

$$p=q=R_t\sin\theta_B=2R_w\sin\theta_B$$

$$\frac{\Delta E}{E}=\sqrt{\omega_D^2+\Delta^2_{\text{geom}}+\Delta^2_{\text{source size}}}\cot\theta_B$$



$$\Delta_{\text{geom}}=\Delta_{\text{source}}$$



$$\Delta_{\text{geom}}=\Delta_{\text{source}}-\frac{x}{R_t}=$$

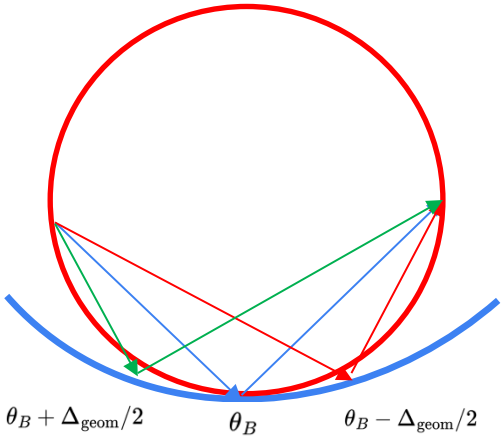
$$\Delta_{\text{source}}-\frac{p\Delta_{\text{source}}/\sin\theta_B}{R_t}=$$

$$\Delta_{\text{source}}\left(1-\frac{p}{R_t\sin\theta_B}\right)$$

Rowland configuration

- cancels out the contribution due to source divergence
- gives the best possible energy resolution

Curved crystals: Johann spectromerer



- H. H. Johann, "Die Erzeugung lichtstarker Röntgenspektren mit Hilfe von Konkavkristallen" ("The production of intense X-ray spectra with the aid of concave crystals"), Zeitschrift für Physik, vol. 69, pp. 185–206, 1931.
- The usual bent crystal in Rowland geometry is also called Johann mounting
- It is an X-ray focusing device where a **flat crystal is simply bent** to a radius Rt.
- Source and focus and placed on the Rowland circle of radius Rt/2
- The crystal touches the Rowland circle of radius at its center. Therefore
  i) the rays are not perfectly converging to the focus,
  ii) only the central ray is incident at Bragg angle, thus reducing transmittivity
- Not efficient for long crystals.
- “Johann error” Expression?

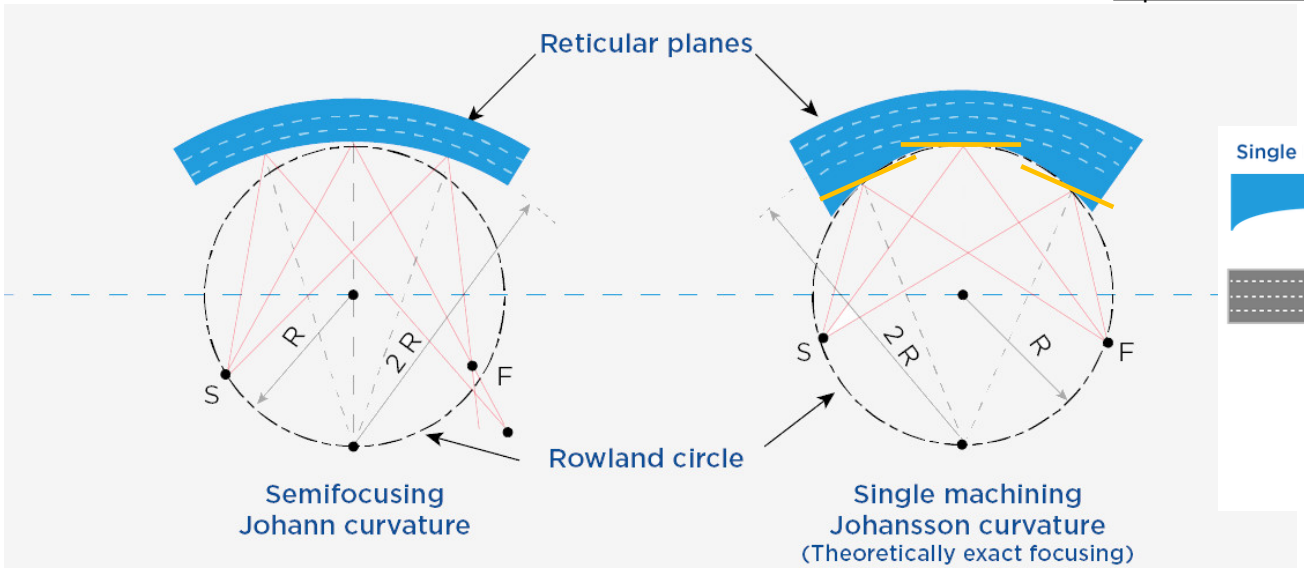
using the Rowland circle radius $f = R_t \sin\theta_B/2$:

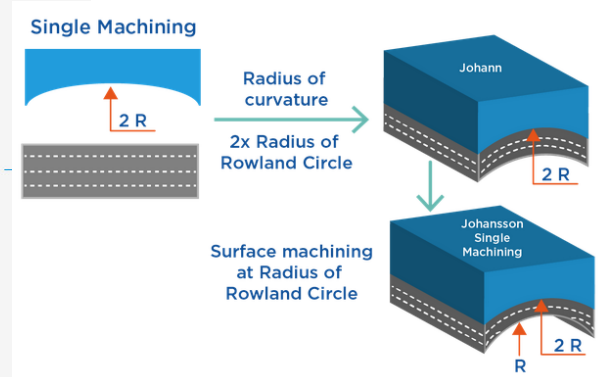
$$\frac{\Delta E}{E} = \frac{L^2}{8f^2}\cot^2\theta_B$$

$$\Delta\theta = \frac{L^2}{8f^2}\cot\theta_B$$

Johansson

https://luxiumsolutions.com/x-ray-analyzers



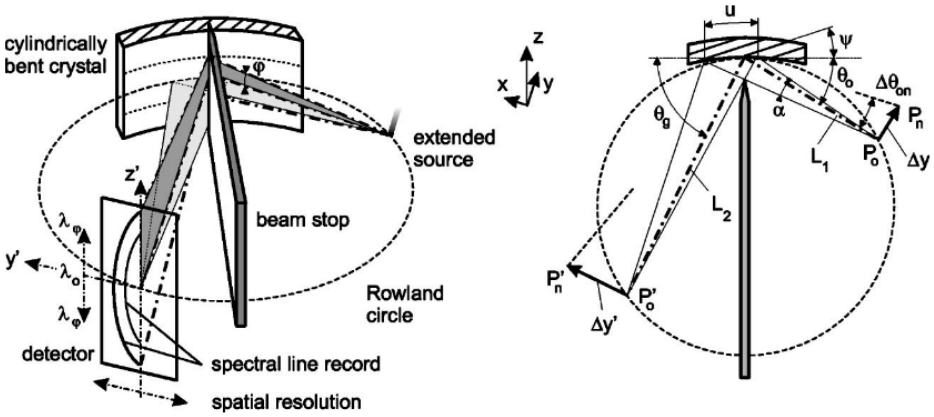


How to solves all the problems?

- The surface is forced to be on the Rowland circle: good focusing
- The Bragg planes are “adjusted” to be always perpendicular to the **angle bisector** between in and out rays: good transmission

Curved crystals: VALID FOR ASYMMETRIC CRYSTALS

https://doi.org/10.1063/1.1805253
Or https://doi.org/10.1039/D3JA00437F



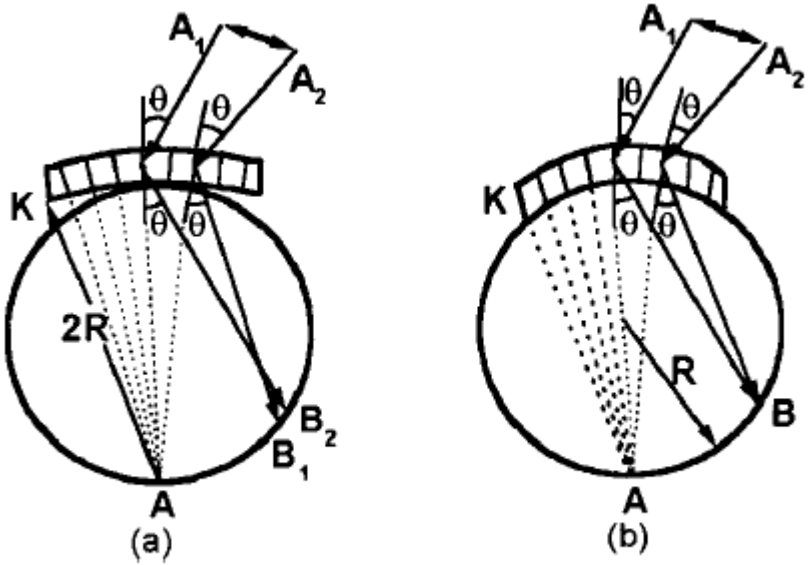
Lens equation

$$\frac{\sin^2\theta_0}{p}+\frac{\sin^2\theta_g}{q}=\frac{\sin\theta_0+\sin\theta_g}{R_h}$$

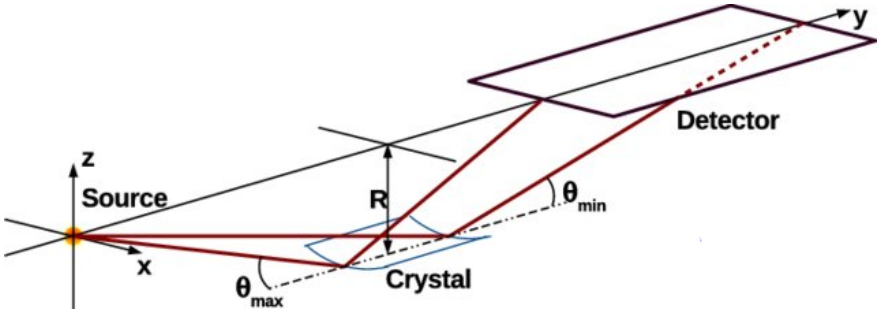
For Rowland configuration

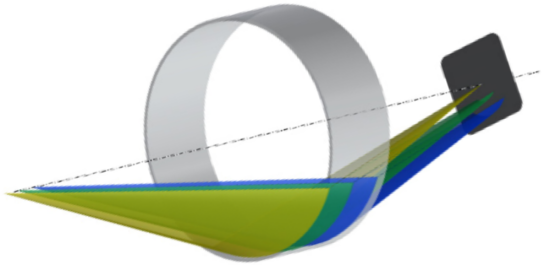
$$\frac{\sin\theta_0}{L_1}+\frac{\sin\theta_g}{L_2}=\frac{2}{R_h}$$

Other Spectrometers



(a) Cauchois-Johann and (b) Cauchois–Johansson optical schemes.

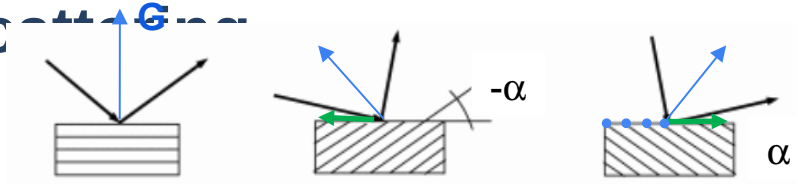




Von Hamos

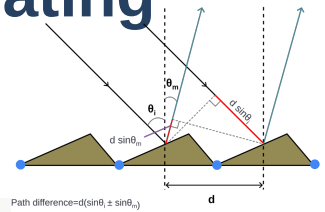
https://doi.org/10.1063/1.1324754

Crystal vs Grating



$$|\mathbf{k}_1| = |\mathbf{k}_2|$$

$$\mathbf{k}_{2,\parallel} = \mathbf{k}_{1,\parallel} + \mathbf{G}_\parallel$$



$$|\mathbf{k}_{out}| = |\mathbf{k}_{in}| = 2\pi/\lambda$$

$$\mathbf{k}_{out,\parallel} = \mathbf{k}_{in,\parallel} + m\,\mathbf{G}$$

$$\hat{\mathbf{k}}_{out} - \hat{\mathbf{k}}_{in} = m\frac{\lambda}{d}\hat{\mathbf{g}}$$

$$k_{in,\parallel} = \frac{2\pi}{\lambda}\sin\theta_{in}$$

$$k_{out,\parallel} = \frac{2\pi}{\lambda}\sin\theta_{out}$$

A crystal acts like a grating with effective period d / m = dhkl / sin α (dispersive)

$$\sin\theta_2 = \sin\theta_1 + \frac{\lambda}{d_{hkl}}\sin\alpha$$

$$\sin\theta_{out} = \sin\theta_{in} + \frac{m\lambda}{d}$$

The difference is that the crystal adds the Bragg wavelength-selection condition on top of the grating direction-change condition.

A symmetric Bragg it is specular like m=0 (non dispersive), but the crystal acts like a **wavelength-selective mirror** —but it has an intrinsic bandpass set by the dynamical diffraction theory.

Typically:

- Grating monochromator: grating + slit
- Crystal monochromator: just crystal
- The same focusing equations (Lens equations)

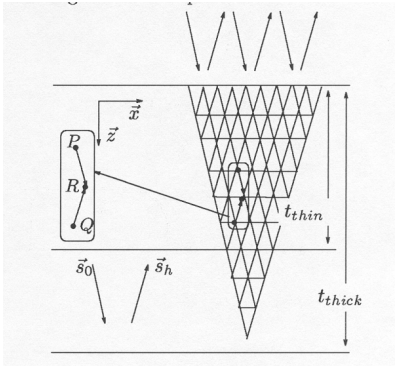
Distorted crystal (reflectivities)

Takagi, Acta Crystal 15 (1962) 1311

Taupin, Bull Soc Fr. Mineral. 84 (1961) 51

$$\frac{\partial}{\partial s_0} D_0(\vec{r}) = -i\pi k P \Psi_{\bar{H}} D_h(\vec{r});$$

$$\frac{\partial}{\partial s_h} D_h(\vec{r}) = -i\pi k P \Psi_H D_0(\vec{r}) + 2\pi i \left\{ k\beta_h - \frac{\partial}{\partial s_h} \left(\vec{h}.\vec{u}(\vec{r}) \right) \right\} D_h(\vec{r}),$$



Analytical solutions of Takagi-Taupin (TT) only for a few cases

Numerical integration of the PDF (2D)

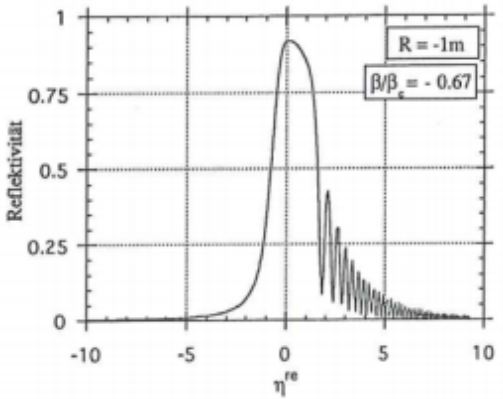
- Regular grids (Finite Different Methods) Authier, Epelboin, Gronkowski, etc.
- Finite Element Methods Ari-Pekka Honkanen (https://doi.org/10.1107/S1600576718001930) Yuhang Wang : (https://doi.org/10.1364/OE.409961)

If deformation *u(z)* (e.g., uniform curvature) the PDEs reduce to a single ODE

$$ic\frac{\mathrm{d}X}{\mathrm{d}\eta} = X^2(1+i\varkappa) - 2X(\eta+ig) + (1+i\varkappa).$$

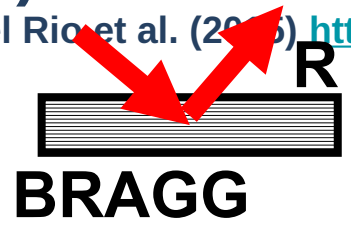
$X = \sqrt{\frac{|\gamma_h|}{\gamma_0}} \sqrt{\frac{\chi_h}{\chi_h}} \frac{D_h}{D_0}$. $c = \mathrm{d}\eta/\mathrm{d}A$.

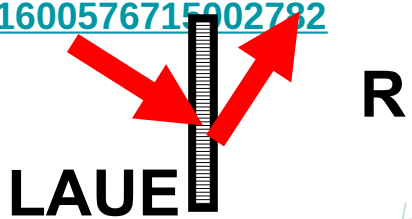
M. Krisch, Ph. Thesis (1993)
Hölzer et al. Cryst. Res. Technol. 33, 555 (1998)

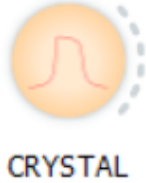


Approximated methods for bent crystals reflectivity (in XOPPY)

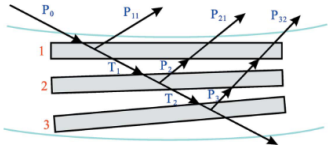
Sanchez del Rio et al. (2015) https://doi.org/10.1107/S1600576715002782



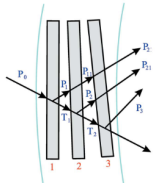


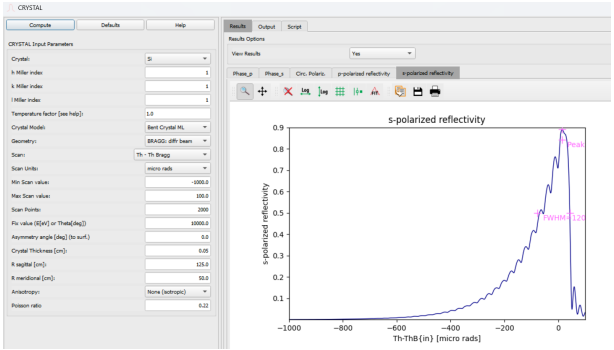


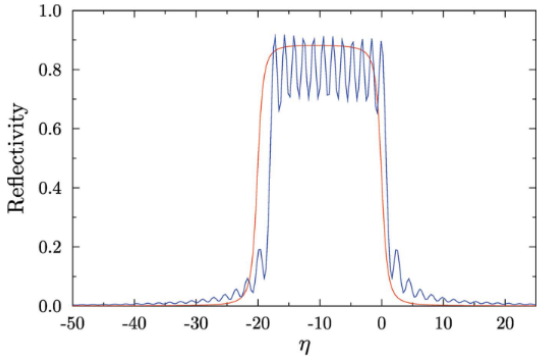
- Multilamellar



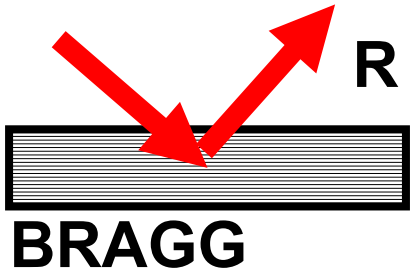
- Multilamellar
- Penning-Polder

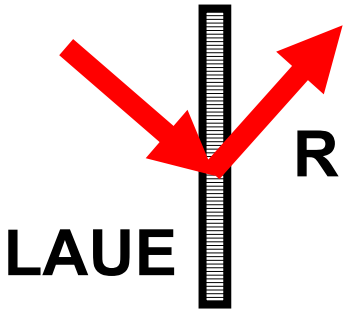


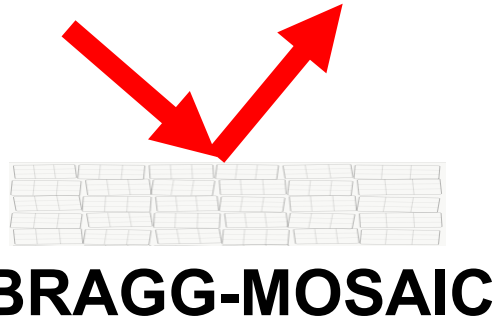


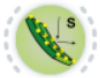


SHADOW4 State-of-the-art



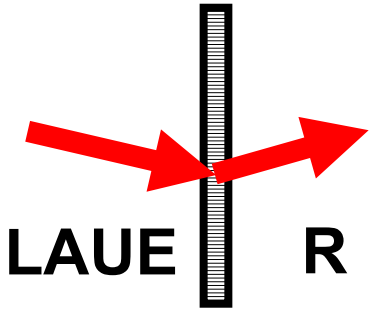




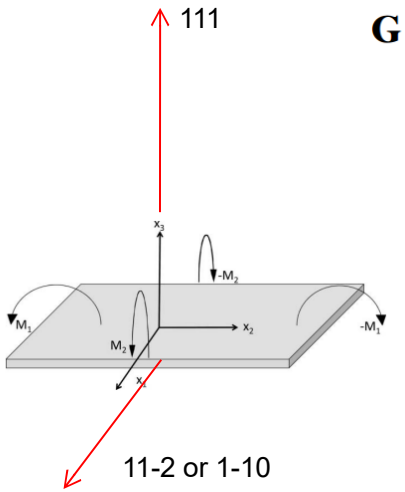
	Curved Undistorted	Curved Distorted		
 Spherical Crystal SHADOW4				
SHADOW3			 Limited applications: Undistorted No dynamical focusing	

Why LAUE crystal focusing is complicated?

LAUE crystals are used for high energy monochromators (θ_B <<)



Effect of crystal anisotropy

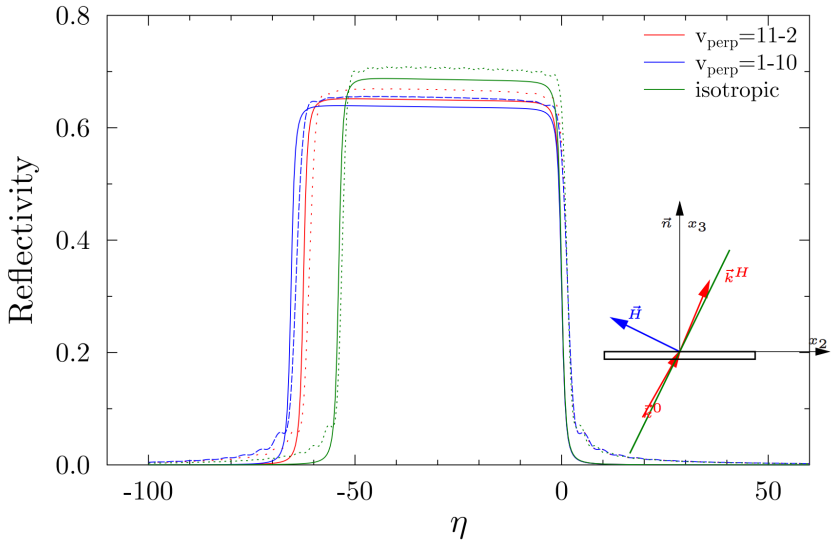


$$\mathbf{G} \equiv \frac{\partial^2(\mathbf{H}\cdot\mathbf{u})}{\partial V_0 \partial V_H}$$

$$\begin{aligned}
u_1 &= \frac{1}{I}[(s_{11}M_1 + s_{12}M_2)x_1x_3 + (s_{51}M_1 + s_{52}M_2)x_3^2/2 \\
&\quad + (s_{61}M_1 + s_{62}M_2)x_2x_3/2], \\
u_2 &= \frac{1}{I}[(s_{21}M_1 + s_{22}M_2)x_2x_3 + (s_{41}M_1 + s_{42}M_2)x_3^2/2 \\
&\quad + (s_{61}M_1 + s_{62}M_2)x_1x_3/2], \\
u_3 &= \frac{1}{2I}[-(s_{11}M_1 + s_{12}M_2)x_1^2 - (s_{21}M_1 + s_{22}M_2)x_2^2 \\
&\quad - (s_{61}M_1 + s_{62}M_2)x_1x_2 + (s_{31}M_1 + s_{32}M_2)x_3^2].
\end{aligned}$$

$$\frac{M_1}{I} = \frac{1}{s_{12}s_{21} - s_{11}s_{22}}\left(\frac{s_{12}}{R_2} - \frac{s_{22}}{R_1}\right),$$
$$\frac{M_2}{I} = \frac{1}{s_{12}s_{21} - s_{11}s_{22}}\left(\frac{s_{21}}{R_1} - \frac{s_{11}}{R_2}\right).$$

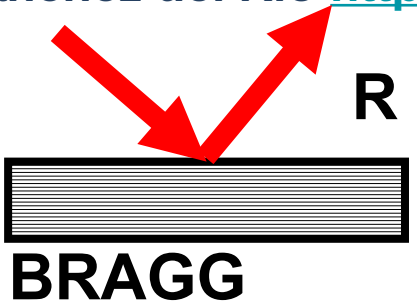
Crystal	Si 111
Photon energy	33170 eV
Crystal thickness	0.07 cm
Meridional Radius	325 cm (convex to the beam)
Asymmetry	$\alpha_X = 296.2°$ ($\alpha = -116.2°$, $\chi = -206.2°$)

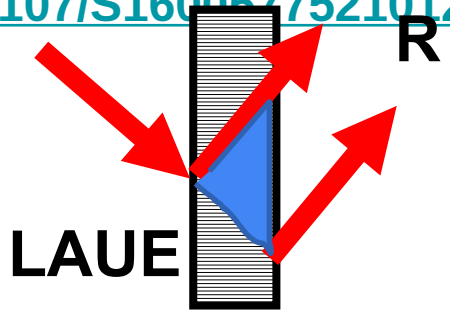


M. Sanchez del Rio, N. Perez-Bocanegra, X. Shi & V. Honkimaki, J. Appl. Cryst. (2015). 48, 477-491

Focusing (Lens equation: monochromatic focus by surface)

Guigay Sanchez del Rio https://doi.org/10.1107/S1600577521012480





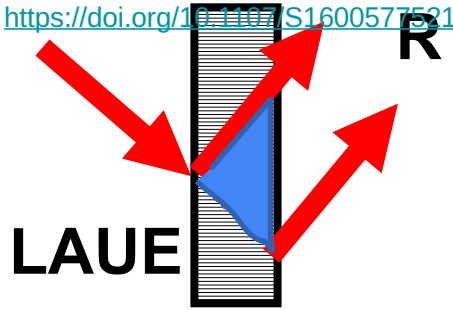
Not working
Bulk effect (Borrmann triangle)

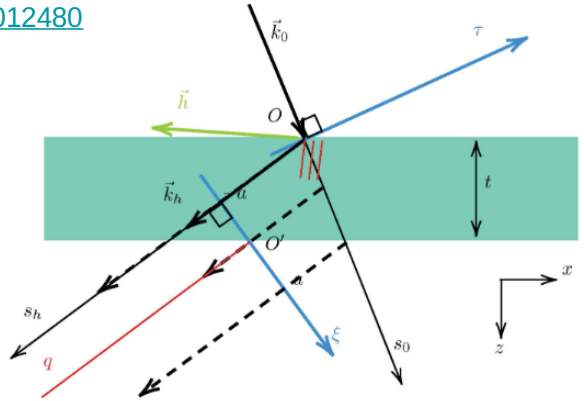
Lens equation:
“surface effect”

$$\frac{|\cos\varphi_h|\cos\varphi_h}{L_h} - \frac{\cos^2\varphi_0}{L_0} = \frac{\cos\varphi_h - \cos\varphi_0}{R_c}$$

Focusing Borrmann triangle (monochromatic)

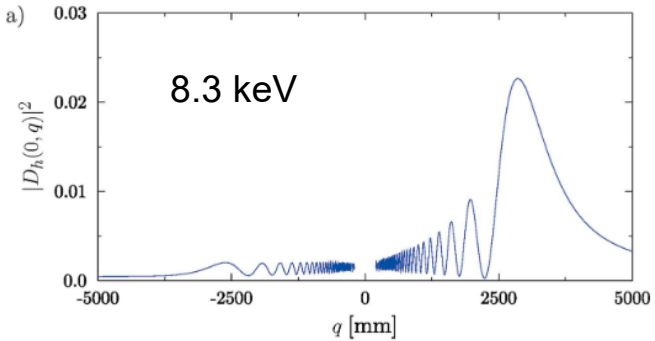
Afanas'ev, A. M. & Kohn, V. G. (1977). *Fiz. Tverd. Tela (Leningrad)*, 19, 1775–1783.
Guigay *et al.* https://doi.org/10.1107%2FS0108767312044601 https://doi.org/10.1107%2FS2053273316006549
https://doi.org/10.1107/S1600577521012480

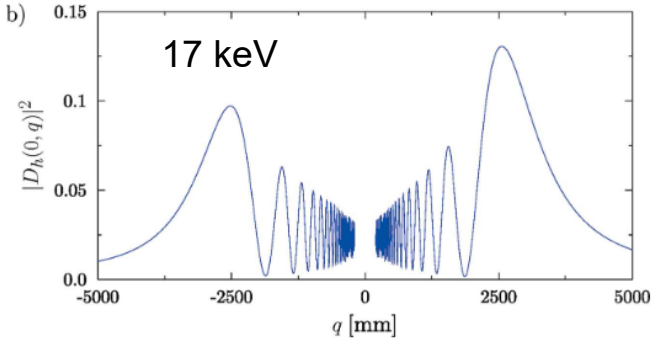




$$D_h(\xi; q) = (\lambda q)^{-1/2} \int_a^a d\xi' \exp\left[ik\frac{(\xi-\xi')^2}{2q}\right] J_0\left[Z(a^2-\xi^2)^{1/2}\right]$$

250 um-thick flat Si111



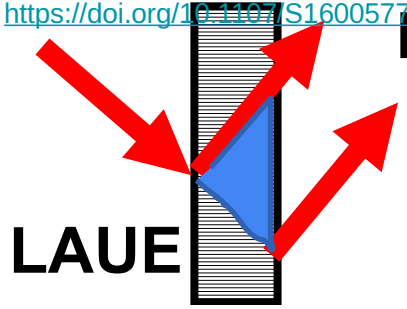


- Plane incident waves gives non-plane reflected waves
- From the quadratic phase we can obtain the focal position
- Not possible to directly implement in ray tracing

Focusing Borrmann triangle

Afanas'ev, A. M. & Kohn, V. G. (1977). *Fiz. Tverd. Tela (Leningrad)*, 19, 1775–1783.

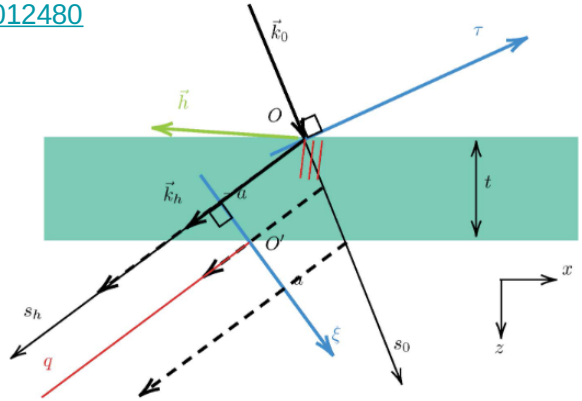
Guigay *et al.* https://doi.org/10.1107%2FS0108767312044601 https://doi.org/10.1107%2FS2053273316006549 https://doi.org/10.1107/S1600577521012480





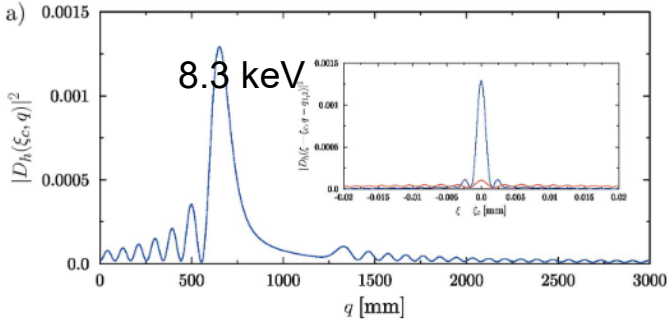
Laue Crystal 1D

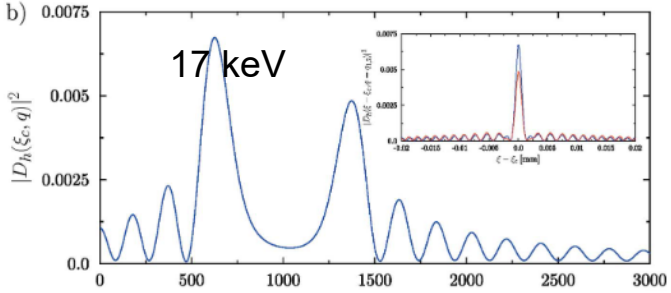
Soon in WOFRY!



$$D_h(\xi, q) = \frac{\exp[ik(\xi^2/2L)]}{(\lambda L)^{1/2}} \times \int_{-a}^{+a} d\eta \exp\left\{\frac{ik}{2}\left[\frac{\eta^2}{L_e} - \eta\left(\frac{2\xi q_e}{qL_e} + \frac{a}{R'}\right)\right]\right\} \times J_0\left[Z(a^2 - \eta^2)^{1/2}\right], \quad (23)$$

250 um-thick symmetric bent Si111



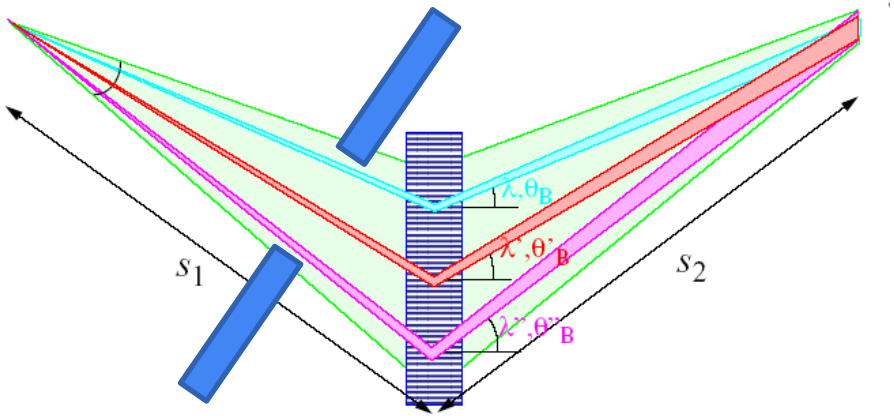


- Spherical Plane incident waves gives non-plane reflected waves
- New lens equation
- Only for symmetric Laue

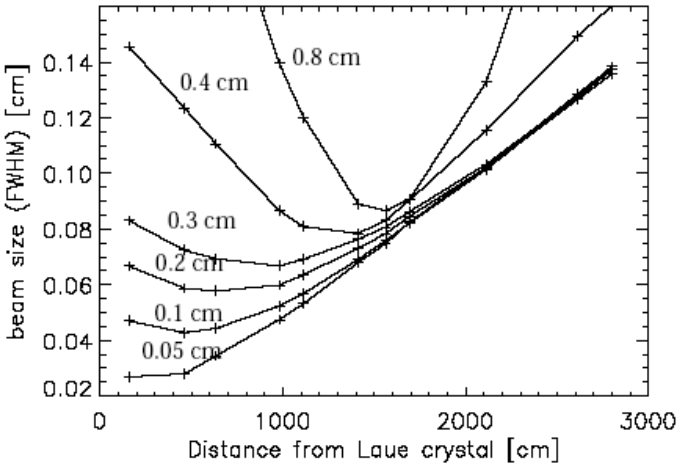
$$\frac{1}{L_h - R_c \cos\theta_B} - \frac{1}{L_0 - R_c \cos\theta_B} = \pm \frac{q_{dyn}}{(R_c \cos\theta_B)^2}$$

Laue: polychromatic focusing effects

- Polychromatic focusing with flat Laue crystals is *fake* focusing in 1:1 geometry.
- Produced by lateral variation of Bragg angle variation
- The monochromatic components are not focusing (lens equation)
- Even neglecting dynamical effects (Borrmann considerations)



M. Sanchez del Rio et al, Rev Sci Instrum, 66 (11)
5148- 5152, Nov 1995

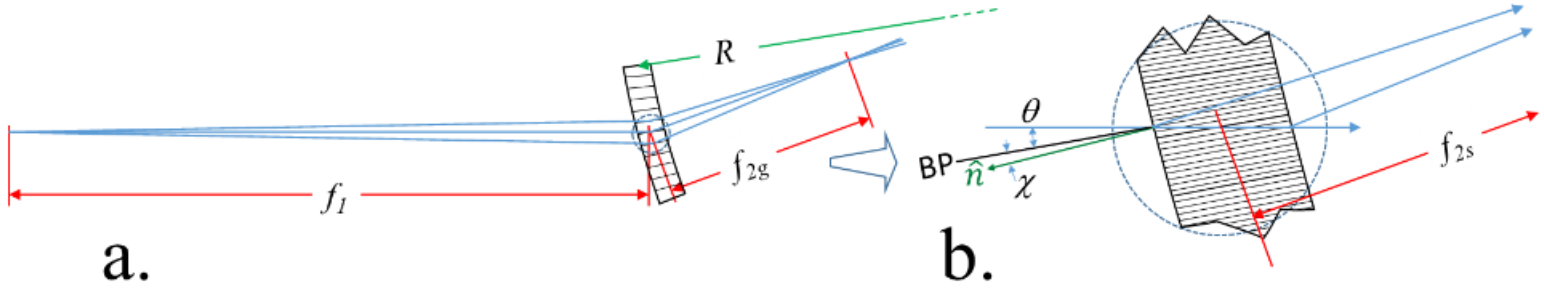


https://doi.org/10.1016/0168-9002(94)91905-4
https://dx.doi.org/10.1063/1.1146141

Laue: polychromatic focusing effects: Magic condition

https://doi.org/10.1107/S1600576720016428
https://doi.org/10.1117/12.2525449
Peng Qi, Xianbo Shi, Nazanin Samadi, Dean Chapman

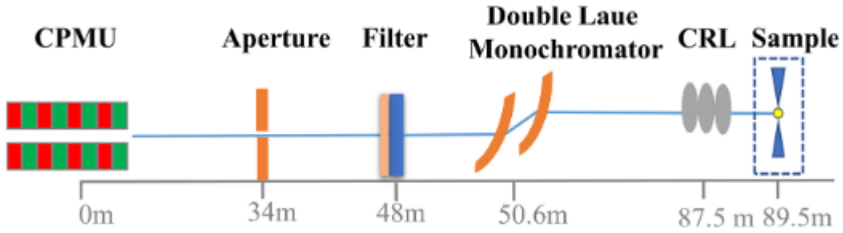
Suppose, for simplicity, that a focusing bent Laue crystal is considered with a polychromatic incident x-ray beam. Similar arguments will apply for a defocusing or virtual focus system. The focal properties of such a crystal is complicated by two effects. One, there is a geometric focus arising from the overall curvature of the lattice in the diffraction plane. Also, each ray that intercepts the crystal will generate a bundle of diffracted rays that will focus (either a real or virtual focus). An example of the geometric focusing arrangement is shown in Fig. 2a and the effect of diffraction on each ray in Fig. 2b.



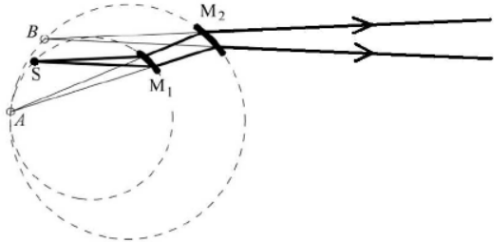
See also: https://doi.org/10.1107/S0021889804023829

Laue-Laue monochromators

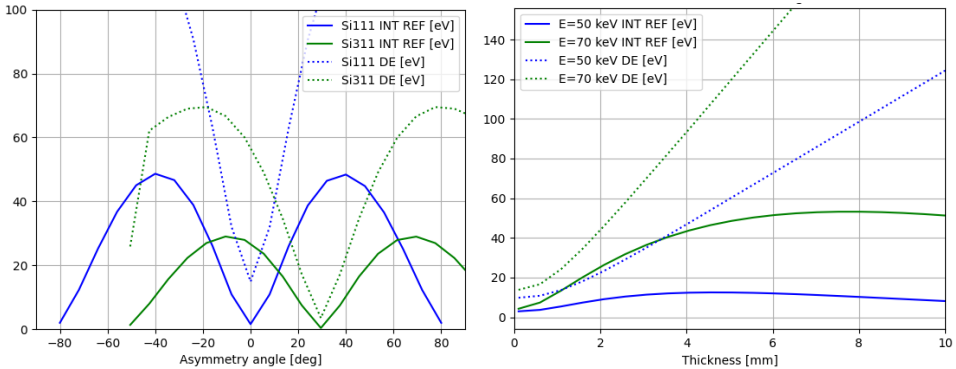
Beijing SRF: Wang et al. (2025) XOPPY+XRT https://doi.org/10.1107/S1600576724011841



ESRF
ID11 future cryogenic LL monochromator
Based on existing ID31 and ID15 monochromators



ID31 XOPPY reflectivity simulations (Penning-Polder)



Thank you!

More slides

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- Ewald (1917) *Zur Begründung der Kristalloptik, Teil III Die Kristalloptik der Röntgenstrahlen* Ann. Phys. 54, 519
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- Takagi (1962) *Dynamical theory of diffraction applicable to crystals with any kind of small distortion* Acta Crystal 15 1311
- Recent books:

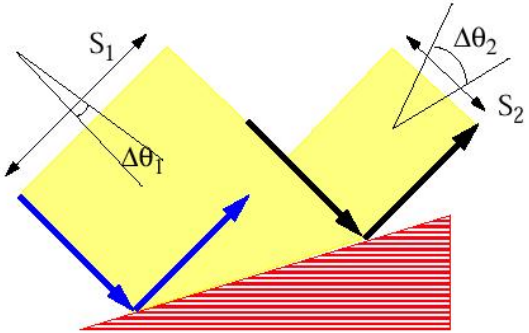
  Pinsker *Dynamical Scattering of X rays in Crystals* (1978).

  Authier *Dynamical Theory of X-Ray Diffraction* (2001)

  Shvyd'ko *X-Ray Optics—High-Energy-Resolution Applications* (2004)
- *crystalpy* Guigay Sanchez del Rio JSR (2024)

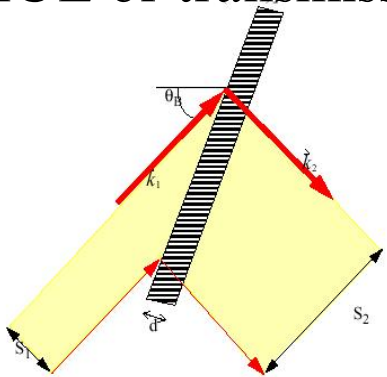
Zachariasen Formalism (undistorted)

BRAGG or reflection

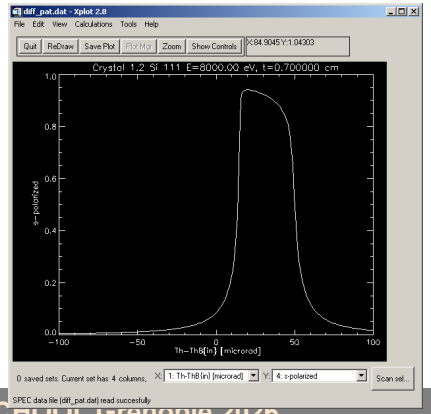


$$r^{\text{Bragg}}(\alpha_Z) \equiv \frac{1}{|b|}\frac{I^H}{I^0} = \frac{1}{|b|}\left|\frac{x_1 x_2 (c_1 - c_2)}{c_2 x_2 - c_1 x_1}\right|^2$$

LAUE or transmission



$$r^{\text{Laue}}(\alpha_Z) \equiv \frac{1}{|b|}\frac{I^H}{I^0} = \frac{1}{|b|}\left|\frac{x_1 x_2 (c_1 - c_2)}{x_2 - x_1}\right|^2$$



$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{-z \pm (qP^2 + z^2)^{1/2}}{P\Psi_{\overline{H}}}$$

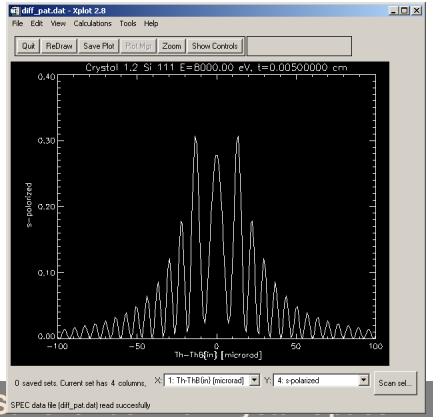
$$z = \frac{1-b}{2}\Psi_0 + \frac{b}{2}\alpha_Z.$$

$$c_1 = \exp(-i\varphi_1 T), \quad c_2 = \exp(-i\varphi_2 T),$$

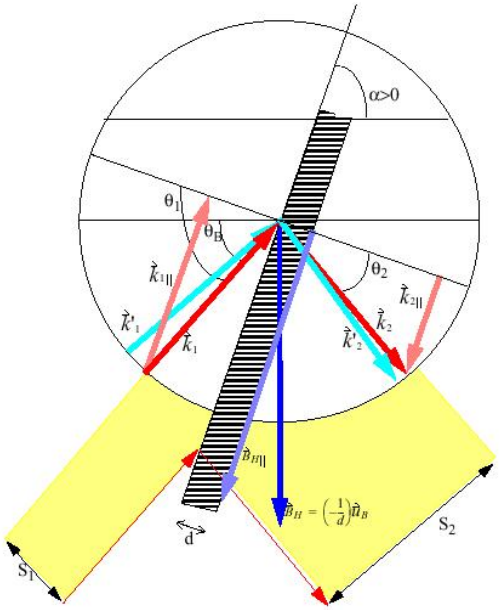
$$\varphi_1 = -\frac{2\pi k^0 \delta_0'}{\gamma_0}, \quad \varphi_2 = -\frac{2\pi k^0 \delta_0''}{\gamma_0},$$

$$\begin{pmatrix} \delta_0' \\ \delta_0'' \end{pmatrix} = \frac{1}{2}\left[\Psi_0 - z \pm (qP^2 + z^2)^{1/2}\right]$$

$$\alpha_Z = \frac{1}{|\mathbf{k}^0|^2}\left(|\mathbf{H}|^2 + 2\mathbf{k}^0 \cdot \mathbf{H}\right)$$



Geometrical model (surface model, no bulk effect)



The change in the direction of any monochromatic beam (not necessarily satisfying the diffraction condition or Laue equation) diffracted by a crystal (Laue or Bragg) can be calculated using (i) elastic scattering in the diffraction process:

$$|\mathbf{k}^0| = |\mathbf{k}^H| = \frac{1}{\lambda} \quad (2)$$

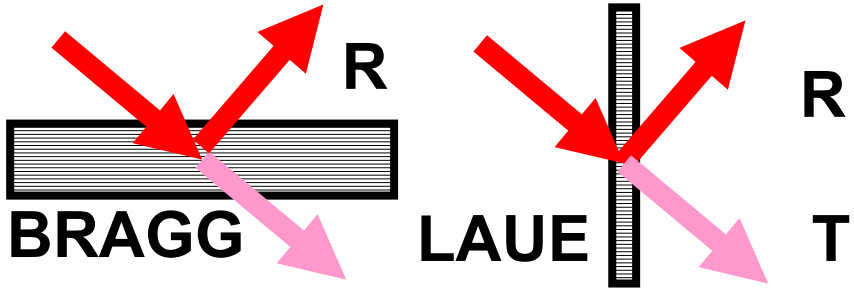
with $\mathbf{k}^{0,H} = (1/\lambda)\mathbf{V}^{0,H}$ and $\mathbf{V}$ a unitary vector; and (ii) the boundary conditions at the crystal surface:

$$\mathbf{k}_{\parallel}^H = \mathbf{k}_{\parallel}^0 + \mathbf{H}_{\parallel}. \quad (3)$$

$$-|\sin\theta_2| = |\sin\theta_1| - \frac{\lambda}{d}\sin\alpha$$

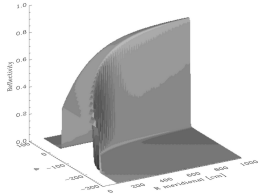
$$\frac{d}{\sin\alpha} = \frac{d_{Grating}}{m}$$

- Plane waves (incident and reflected)
- Common intersection point on surface (no depth)
- Symmetric Bragg is non-dispersive (specular).
  All other cases are dispersive (non-specular, like gratings or prisms)

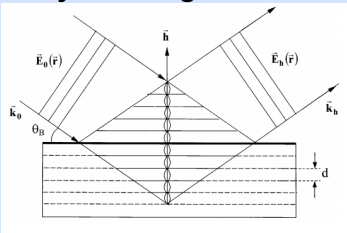


Takagi, Acta Crystal 15 (1962) 1311

Taupin, Bull Soc Fr. Mineral. 84 (1961) 51



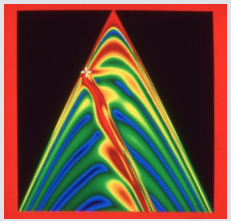
X-ray standing waves



The fluorescence signal is directly proportional to the internal wave field intensity at the atom position . The position of the fluorescing atom with respect to the diffraction plane can be determined.

Batterman and Cole. Rev. Mod. Phys. 36 (1964) 681

X-ray topography



Simulation of dislocations, stacking faults, cracks, etc.

Y. Epelboin, Progress in Crystal Growth and Characterization, 14, (1987)465

Grazing incidence diffraction

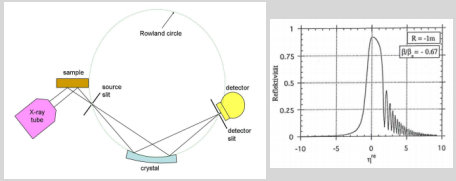


The angles involved are close to the critical angle. Fresnel specular theory predicts an evanescent wave, which must be treated by the dynamical theory.

S. Stepanov
http://x-server.gmca.aps.anl.gov/

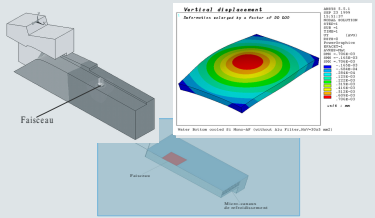
Using Dynamical theory in x-ray INSTRUMENTS

Curved spectrometers, monochromators and analyzers



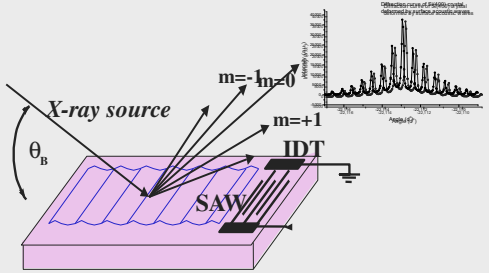
M. Krisch, Ph. Thesis (1993)
Hölzer et al. Cryst. Res. Technol. 33, 555 (1998)
Honkanen et al. J. Synchrotron Rad. 21, 104 (2014)
M. Sanchez del Rio et al. J. Synchrotron Rad.48, 477 (2015)

Thermally deformed monochromator



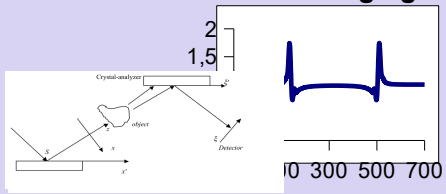
Mocella et al. Nucl. Inst. Meth. in Phys. Res. A 467, 414 (2001).
Mocella et al. J. Appl. Cryst. 36, 129 (2003)
Mocella et al. Proc. SPIE 9510 (2015)
Honkanen, et al. J. Appl. Cryst. 49, 1284 (2016)

Surface acoustic waves



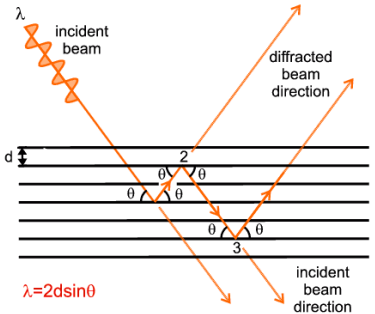
Mocella & Epelboin J. Appl. Cryst. 32, 154, 1999
Touculou et al . J. Appl. Phys. 97, 113505, 2005

Monochromator+Analyzer in Diffraction Enhanced Imaging



Bravin et al.
J. Appl. Phys. 97, 113505 (2005)

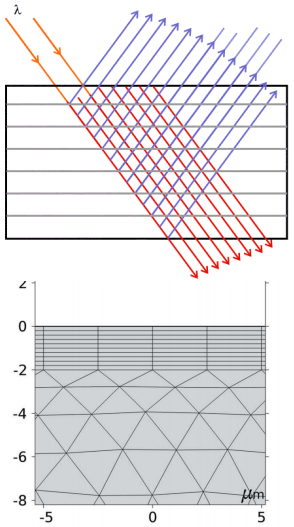
TAKAGI-TAUPIN EQUATIONS

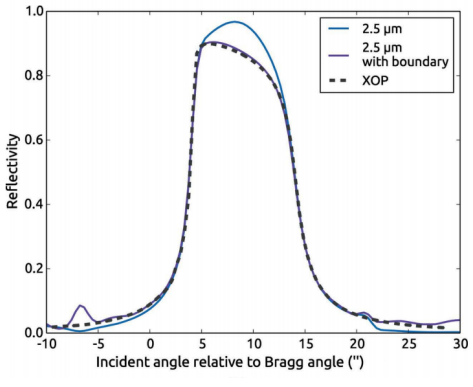


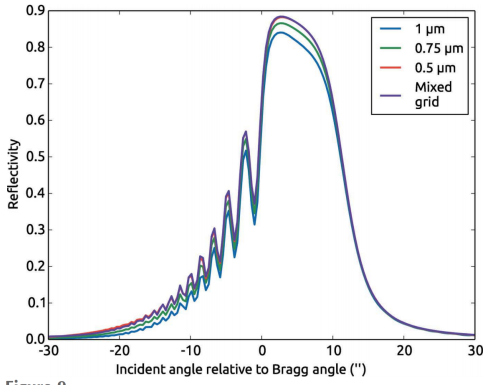
$$\partial_0 D_0 = ic_{\bar{h}} D_h,$$

$$\partial_h D_h = i\left[\beta + \partial_h(\mathbf{h}\cdot\mathbf{u})\right]D_h + ic_h D_0$$

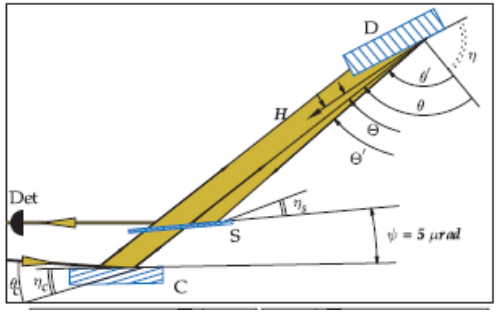
$$c_0 = k\chi_0/2,\ c_{\bar{h},h} = kC\chi_{\bar{h},h}/2,\ \beta = k - k_h \simeq k\Delta\theta\sin 2\theta_B$$







4) Extreme asymmetry (backscattering) produces a rainbow effect Shvyd'ko 2006 PRL 97, 235502



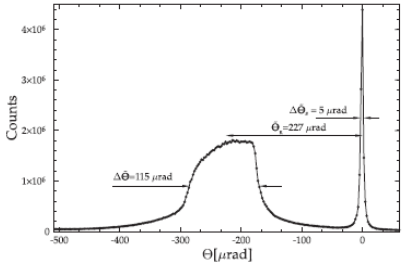
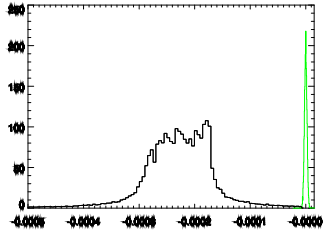
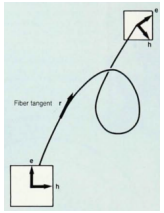
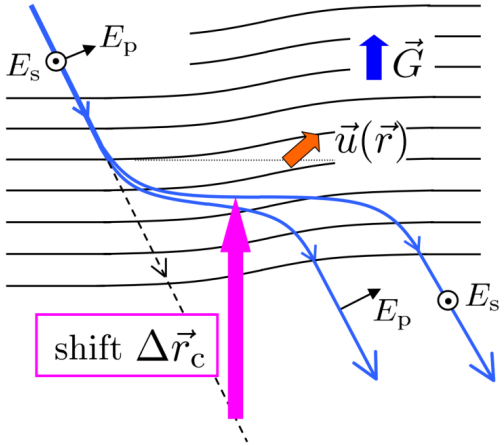


FIG. 4. Bragg reflectivity from crystal D of 9.1315 keV x rays, exactly backwards, as a function of the angle of incidence $\Theta = \pi/2 - \theta$ to the reflecting atomic planes.

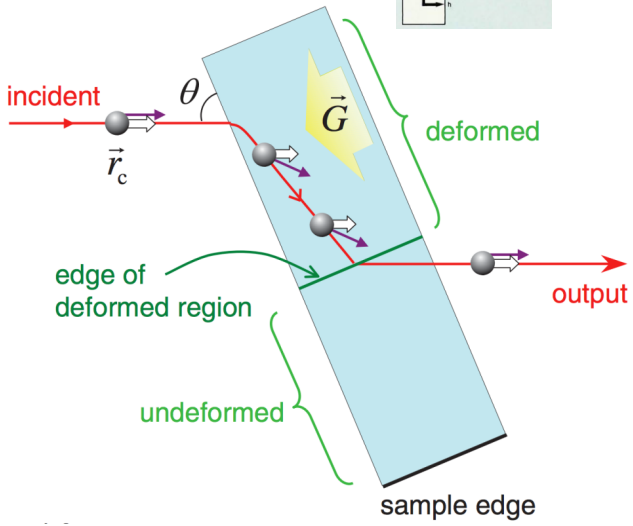


Berry, M.V. Proc. R. Soc. London, Ser. A392, 45 (1984)
Berry, M.V. 1990 Physics Today, 43, 12 34-40



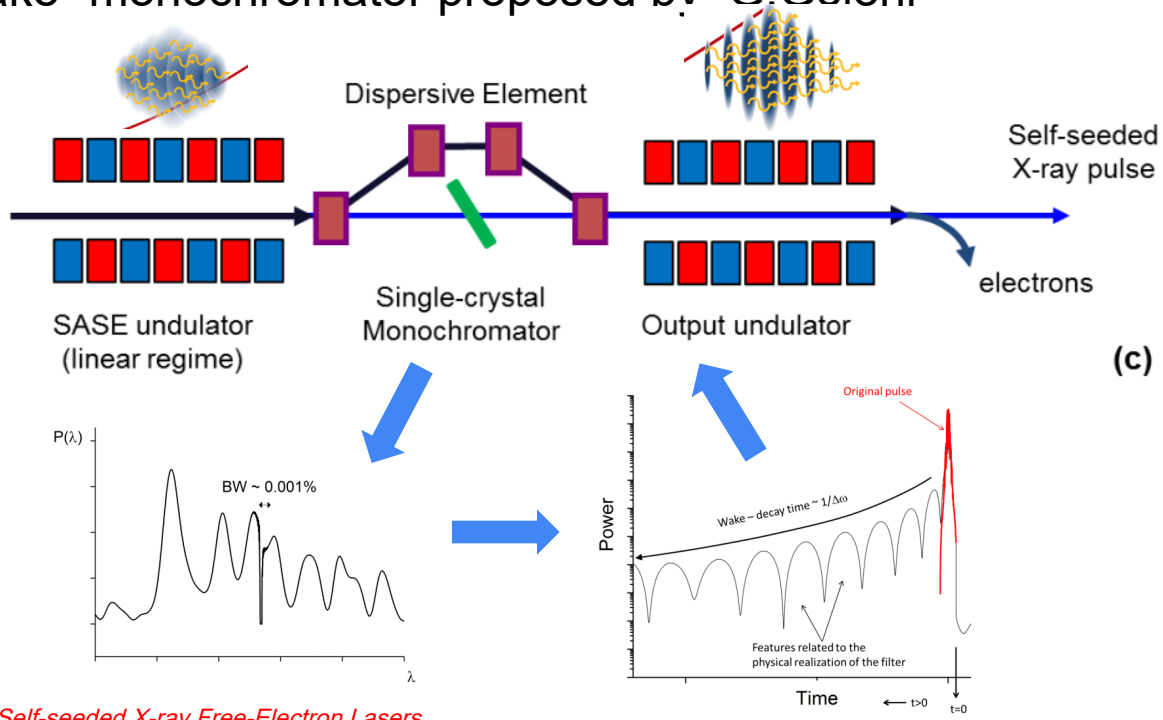


K Sawada, S Murakami, N Nagaosa
Phys. Rev. Lett. 96, 154802 (2006)



Y Kohmura, K Sawada, T Ishikawa
Phys. Rev. Lett. 104, 244801 (2010)

- Time-dependence diffraction was studied in the 1990s
- “wake” monochromator proposed by G.Geloni

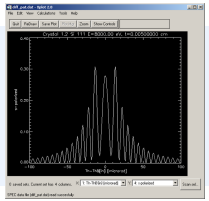


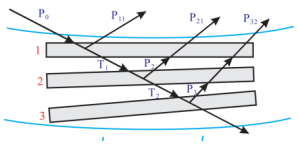
G. Geloni *Self-seeded X-ray Free-Electron Lasers*
https://pubdb.xfel.eu/record/292720/files/RESUB_self_final%20-%20POST_REFEREE.pdf?subformat=pdfa
Y. Shvyd’ko and R. Lindberg, *Spatiotemporal response of crystals in x-ray Bragg diffraction*, Phys. Rev. ST Accel. Beams 15, 100702 (2012) dx.doi.org/10.1103/PhysRevSTAB.15.100702

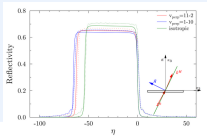
Hierarchy of dynamical theories of x-ray diffraction for deformed and perfect crystals

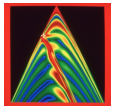
J Härtwig (2001 J. Phys. D 34 A70)

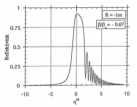
- 1st approximation: dynamical diffraction theory for the perfect crystal
  - e.g, flat monochromators, curved monochromators and spectrometers locally “undistorted”, standing waves, etc. XOP, SHADOW
- 2nd approximation: local applications of the dynamical theory for perfect crystals to distorted ones
  - e.g, curved crystals: Multilamellar method (XOP/XCRYSTAL)
- 3rd approximation: geometrical optics or eikonal theory
  - e.g, curved Laue crystals: Penning-Polder method (XOP/XCRYSTAL)
- 4th approximation: wave optics or Takagi theory
  - 2D numerical integration: Topography codes adapted to user needs.
  - 1D numerical integration: curved spectrometers, monochromators, analyzers
- 5th approximation: wave optics or Takagi theory for many waves (white beams, curved surfaces, incoherent











TO DO